

FAIR: A Design Theory for Artificial Intelligence Fairness

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Abstract. Artificial intelligence (AI)-automated decision systems encounter persistent, interdependent, and dynamic fairness tensions that traditional one-off interventions cannot resolve. Because these tensions persist due to interdependence and dynamic interaction, organizations require both a theory of the problem to explain their persistence and a theory of the solution to prescribe how they can be managed. Our design theory, **FAIR** (Fairness Adaptation through AI-augmented Responsiveness), provides a theory of the problem by reframing AI fairness as a sociotechnical paradox constituted within AI artifacts that automate decision tasks, through interdependent organizational, technical, and governance choices and their interaction with regulatory mandates and societal norms. Synthesizing four fairness perspectives (Ethics, Organizational Justice, Economic Fairness, and Rawlsian Justice), we identify three metatheoretical dimensions (principles, goals, foci) and show that the interdependence within and among these dimensions is the root, endogenous source that constitutes paradoxical fairness tensions. Building on this diagnosis, FAIR provides a theory of the solution by specifying an organizational capability grounded in three design foundations. First, the paradox lens motivates iterative adaptive cycles (Surfacing and Resolving) to continually surface and resolve AI fairness tensions. Second, design science in information systems and computer science distinguishes AI artifacts (the “what”) from the actors (the “who”) responsible for adapting them, establishing the basis for complementary human–AI agent collaboration in the adaptive cycles: AI agents execute monitoring to surface and refinement to resolve tensions, whereas human agents specify objectives, adjudicate trade-offs, and exercise contextual judgment and oversight. Third, the managing-with-AI literature informs how this human–AI agent collaboration should be governed. These foundations yield two reinforcing mechanisms: (i) artifact-level adaptation, achieved through structured human–AI agent collaboration, within and across the layers of the AI decision pipeline—Representation (data), Learning (model), and Calibration (decision); and (ii) portfolio-level, risk-tiered federated governance that structures how human–AI agent collaboration scales across tasks and artifacts, balancing process standardization with configuration choices and human control with AI autonomy based on task risk. Enabled by organizational “fairness complements”—namely, human skills to work with AI agents and structured stakeholder feedback—this sociotechnical design provides organizations with a sustained capability to harmonize global coherence and local flexibility in the responsive adaptation of AI fairness.

Keywords: AI fairness, fairness tensions, paradox lens, design theory, AI artifacts, AI agents, human–AI collaboration, AI governance, AI-automated decision-making

1. Introduction

Organizations are increasingly using artificial intelligence (AI)-automated decision-making in high-stakes domains—health care, finance, criminal justice, education, and hiring—because of its accuracy, standardization, speed, and scalability. Alongside this accelerating deployment, mounting evidence reveals unfair AI decisions across contexts—from inequitable organ transplant prioritization (Obermeyer et al., 2019) to racially disparate mortgage-lending outcomes (Bartlett et al., 2022), among others. When AI systems perpetuate historical disadvantages or introduce new harms, pushback from stakeholders erodes public trust, triggers legal and regulatory backlash, and undermines organizational legitimacy, compromising the long-term viability of organizations. Among these ethical issues, we foreground fairness because it concerns the distribution of benefits and risks from algorithmic decisions (Karunakaran et al., 2025) and inevitably interacts with predictive accuracy, operational efficiency, and economic goals (Cowgill & Tucker, 2020; Fu et al., 2021).

Organizations must treat AI fairness as a core sociotechnical design objective. Building on prior accounts, fairness is an integrative construct spanning procedural, interactional, and outcome dimensions—covering due process, respectful treatment, and equitable results (e.g., Binns, 2018; Colquitt, 2001). It operates across foci—individual, group, and societal—so judgments at one level can conflict with judgments at another. Divergent stakeholder expectations compound these multidimensional and multilevel considerations. Stakeholders may foreground different layers in the AI decision pipeline—from inputs to processes to decisions—and apply distinct lenses at each; for example, bias-free data (inputs), transparency (processes), and equitable allocation (decisions). What one stakeholder deems fair may strike another as unfair, producing conflicts that ripple through the data, model, and decision layers of an AI system. Thus, AI fairness is not merely a technical attribute; it is a sociotechnical design concern, continually reshaped by interactions among institutional policies, historical inequities, stakeholder expectations, and societal norms, rendering it dynamic and contested (Cowgill & Tucker, 2020; Mehrabi et al., 2021).

In our review of the AI fairness literature, including an in-depth synthesis of the information systems (IS) literature (see Appendix for details), we identify four interconnected types of AI fairness tensions: (1) *constraints*, (2) *trade-offs*, (3) *incompatibilities*, and (4) *dilemmas*. Constraints arise from regulations (e.g., the General Data Protection Regulation [GDPR], Health Insurance Portability and Accountability Act [HIPAA]) or from organizational objectives. Trade-offs arise (i) among competing fairness definitions—deciding which to prioritize and to what extent (e.g., Binns, 2018; Chouldechova, 2017; Kleinberg et al., 2017; Mehrabi et al., 2021)—and (ii) between fairness and performance objectives, including predictive accuracy (Hurlin et al., 2026) and operational and economic goals (Bichler et al., 2021). Incompatibilities emerge from contradictions among ethical frameworks (e.g., consequentialism vs. deontology) materially encoded in AI designs (Kane et al., 2021). Dilemmas create “no-win” scenarios—for example, removing sensitive data to protect privacy can obscure disparity detection, whereas retaining that data increases the risk of misuse and discrimination (Xu & Zhang, 2021). Importantly, while these tensions are analytically distinct and may be individually addressed or resolved at a point in time, their interdependence and dynamic interaction render AI fairness tensions persistent within sociotechnical systems, thereby defying one-off or purely reactive solutions (Schad et al., 2016).

Real-world examples show how fairness tensions manifest in AI systems that automate decisions (Veale & Binns, 2017; Wachter & Mittelstadt, 2019). Organ-transplant algorithms can encode racial disparities (Obermeyer et al., 2019), predictive policing models highlight fairness–predictive accuracy trade-offs (Almasoud & Idowu, 2025), hiring algorithms illustrate fairness definitional incompatibilities (Dastin, 2022), and educational placement algorithms underscore dilemmas for vulnerable students (Gillborn et al., 2023). These cases show that fairness tensions are not solely about biased input data but also about how fairness dimensions (e.g., procedural vs. outcome), foci (e.g., individual vs. group), and stakeholder expectations are encoded into predictive models and decision policies. Because fairness is continually reshaped by ethical, organizational, and regulatory forces, approaches relying on static constraints or reactive patches prove inadequate (Dolata et al., 2022; Rana et al., 2022; Robert et al., 2020).

Based on our review of the AI fairness literature and organizational practice, we identify two key needs to advance the theory and practice of AI fairness. First, we require “a theory of the problem” that explains *why* the fairness tensions persist rather than merely documenting that they exist. Second, we require a “theory of the solution” that prescribes *how* to manage the persistent and multilevel nature of AI fairness as a sociotechnical design challenge, grounded in its interdependencies and dynamics. This call for a practical, multilevel design is echoed in recent scholarship on the related, but distinct, problem of workplace inequality (Karunakaran et al., 2025) and the broader ethics of advanced AI assistants (Gabriel et al., 2024). For instance, Karunakaran et al. (2025) called for “multilevel . . . organizational interventions” to mitigate what was diagnosed as “inequality cascades”—harms that

are “compounding and cumulative” across the AI life cycle. Notably, in their review of the “normative” (ethics) literature, they identified the *exact* gap we address: a “lack of focus on how stated normative ideals might be realized in practice” (p. 13).

Accordingly, we develop a design theory on the sociotechnical mechanisms for organizations to manage persistent, interdependent, and dynamically evolving AI fairness tensions. Our scope is AI systems that automate decision tasks in organizations; when paired with conversational or generative-AI front ends, our theorization applies to the decision module.

We propose **FAIR** (Fairness Adaptation through AI-augmented Responsiveness)—a design and action theory (Gregor, 2006; Gregor & Jones, 2007) that reframes fairness as an adaptive, multidimensional organizational capability rather than a fixed attribute. As a design theory, FAIR first provides a theory of the problem by conceptualizing AI fairness as a sociotechnical paradox constituted within AI artifacts that automate decision tasks. We synthesize four fairness perspectives (Ethics, Organizational Justice, Economic Fairness, and Rawlsian Justice) to reveal three metatheoretical dimensions (principles, goals, and foci). We posit that the interdependence within and among these dimensions serves as the root, endogenous source that constitutes the paradoxical nature of fairness tensions, thereby establishing the diagnosis that motivates a new sociotechnical design.

FAIR’s prescriptive design—the theory of the solution—builds on that diagnosis. Informed by the paradox lens (Raisch et al., 2018; Schad et al., 2016; Smith & Lewis, 2011), its artifact-level process frames fairness adaptation as continuous, iterative Surfacing and Resolving cycles. This dynamic oscillates between divergence and convergence of tensions, and clarifies why responsive, ongoing adaptation is necessary, rather than relying on static or reactive fixes.

Informed by IS design science (Abbasi et al., 2024; Hevner et al., 2004) and computer science (Russell & Norvig, 2009), this process involves two distinct constructs. First is the “*what*”: *the AI artifact*, which we conceptualize as a modular pipeline—Representation (data), Learning (model), and Calibration (decision) (Caton & Haas, 2024; Mehrabi et al., 2021)—that executes the decision task. Second is the “*who*”: *the agents* responsible for adapting the artifact.

These elements form FAIR’s dual-mechanism sociotechnical design. Mechanism 1 is artifact-level adaptation, in which AI agents (who monitor, analyze, and propose refinements) and human agents (who specify objectives, adjudicate trade-offs, and exercise ethical judgment, contextual knowledge, and oversight) jointly execute the Surfacing and Resolving cycles. Mechanism 2, informed by the managing-with-AI literature (Asatiani et al., 2021; Grote et al., 2024), is portfolio-level governance of decision tasks. This risk-tiered federated governance defines the collaboration protocol by balancing global process standards with configurable choices and calibrating AI agent autonomy versus human control based on task risk. Enabled by organizational “fairness complements”—namely, human skills to work with AI agents and structured stakeholder feedback on fairness experiences—this design provides the capability to ensure global coherence while allowing for local flexibility in the responsive fairness adaptation of AI artifacts that automate decision-making in organizations.

The remainder of the paper elaborates the theory of the problem, formalizes the artifact-level and portfolio-level requirements implied by that diagnosis, develops FAIR’s design-theoretical solution centered on human–AI agent collaboration, operationalizes its artifact-level adaptation and portfolio-level governance mechanisms, and concludes with theoretical and practical implications.

2. The Theory of the Problem: AI Fairness as a Sociotechnical Paradox

In this section we establish the theory of the problem—FAIR’s diagnosis of the AI fairness challenge. We first ground the problem in the paradox lens, arguing that AI fairness tensions are persistent, interdependent, and dynamic. We then review four theoretical fairness lenses (Ethics, Organizational Justice, Economic Fairness, and Rawlsian Justice) to diagnose the sources of these tensions. Finally, we synthesize these lenses to identify three metatheoretical dimensions and posit that the interdependence within and among these dimensions is the root, endogenous source that constitutes fairness tensions as a sociotechnical paradox. An emergency-department (ED) vignette illustrates the ideas.

2.1. Paradoxical Nature of Fairness Tensions

Fairness tensions in AI artifacts that automate decision-making for a given decision task and organizational context are persistent, interdependent, and dynamic. The multifaceted nature of these tensions is apparent in our synthesis of IS research (see Appendix), which identifies four distinct yet interconnected types: constraints, trade-

offs, incompatibilities, and dilemmas. Collectively, AI fairness tensions exhibit paradoxical characteristics due to two reinforcing dynamics: endogenously, ongoing contestation among diverse stakeholders over how fairness is operationalized within AI artifacts; and exogenously, continual influence from regulatory mandates and shifting societal norms (Smith & Lewis, 2011). For example, during a respiratory-virus surge, ED triage must balance throughput (lives saved) with distributional equity. As stakeholder expectations evolve and external conditions change, these tensions repeatedly resurface rather than converge to a stable resolution. Taken together, these features distinguish AI fairness tensions from resolvable trade-offs, positioning them as enduring sociotechnical paradoxes rather than transient design problems.

2.2. Theoretical Perspectives to Diagnose the AI Fairness Problem

2.2.1. Ethics: Incompatibilities and Dilemmas

Core Premise. Ethical perspectives on fairness center on two competing frameworks: deontological and consequentialism (Floridi et al., 2018). Deontology emphasizes moral duties, rights, and procedural integrity—such as privacy, autonomy, and transparency—regardless of outcomes. Consequentialism evaluates fairness based on outcomes even if doing so involves trade-offs or compromises individual rights. These opposing assumptions often lead to ethical dilemmas where organizations must choose between equally problematic alternatives.

Foci. Deontology focuses on protecting individual rights, transparency, and accountability, whereas consequentialism prioritizes societal-level outcomes.

Tensions and Assumptions. Ethical tensions in AI fairness emerge when deontological principles conflict with consequentialist outcome optimizations. These can evolve into ethical dilemmas in which every option has some cost—for example, removing sensitive attributes to safeguard privacy can inadvertently worsen disparities (Xu & Zhang, 2021).

Diagnosis of the AI Fairness Problem. This lens reveals a fundamental incompatibility in AI design: a “fair” system from a consequentialist (utility) view can be “unfair” from a deontological (rights) view that results in incompatibilities and even manifests as dilemmas (Rhue, 2023; Wu et al., 2021). In ED triage, adding a neighborhood-deprivation index can improve risk calibration and total utility (consequentialist) but clashes with rights-based limits on using sensitive attributes (deontological), requiring adjudication based on contextual factors.

2.2.2. Organizational Justice: Procedural Constraints and Trade-Offs

Core Premise. Organizational justice focuses on distributive, procedural, and interactional fairness within institutions (Colquitt, 2001; Colquitt & Rodell, 2011).

Foci. It applies to stakeholders at both individual and group levels, emphasizing equitable treatment and transparent decision processes.

Tensions and Assumptions. Two tensions arise: (1) demands for uniform, transparent criteria that can constrain context-specific interventions, and (2) equity-oriented exceptions that may breach procedural consistency and undermine trust (Cowgill & Tucker, 2020; Raji et al., 2020). Classic organizational justice work also presumes relatively stable institutional structures.

Diagnosis of the AI Fairness Problem. This lens highlights the tension between *consistency* and *context*, in that a “fair” procedure (like a uniform rule) can conflict with the need for equitable, context-specific exceptions, undermining stakeholder trust (Robert et al., 2020). Algorithm bias can further undermine these aims, creating unequal benefits and burdens (Binns, 2018; Kordzadeh & Ghasemaghaci, 2022). In ED triage, published thresholds—that is, risk-score cutoffs for admit/observation/discharge or escalation—promote due process, while bounded, logged overrides for rare cases can advance equity without undermining procedural regularity.

2.2.3. Economic Fairness: Collective Efficiency vs. Structural Inequity Trade-Offs

Core Premise. Under the principle of Pareto efficiency, equity improvements are allowed only if they do not reduce collective welfare (Fu et al., 2021; Hochman & Rodgers, 1969).

Foci. Its emphasis is on system-level efficiency rather than individual rights.

Tensions and Assumptions. Efficiency-driven optimizations can conflict with organizational justice norms or privacy concerns (Kane et al., 2021; Rhue, 2023), and prioritizing collective efficiency may exacerbate inequities or marginalize disadvantaged groups (Teodorescu et al., 2021). Pareto efficiency is distribution-blind and does not correct historical disparities, so explicit equity interventions are often required.

Diagnosis of the AI Fairness Problem. This lens reveals a persistent trade-off between *collective efficiency* and *distributive equity*. A distribution-blind design that maximizes welfare (Pareto efficiency) can reinforce or exacerbate structural inequities. In ED triage, a throughput-maximizing bed-allocation policy can raise aggregate welfare yet systematically lengthen waits for underserved groups unless equity-aware constraints are imposed.

2.2.4. Rawlsian Justice: Equity–Efficiency Trade-Offs

Core Premise. Rawlsian justice centers equity for the least advantaged: inequalities are acceptable only if they improve the position of those worst off (the “difference principle”), consistent with basic liberties and fair equality of opportunity (Rawls, 1971).

Foci. It applies at group or societal levels, emphasizing redistributive interventions that favor underserved communities (Li et al., 2025; Lu et al., 2023).

Tensions and Assumptions. Equity-first commitments can depress short-run efficiency or predictive performance and may conflict with procedural neutrality or operational constraints. Under scarcity, equity measures can escalate into dilemmas when honoring them risks undermining service capacity or organizational viability.

Diagnosis of the AI Fairness Problem. This lens reveals a direct equity-efficiency conflict. It posits that prioritizing the least advantaged (the “difference principle”) can depress overall performance, a trade-off that intensifies under scarcity. In ED triage, prioritizing patients from high-deprivation areas advances the difference principle but can slow throughput—an equity–efficiency trade-off that can harden into a dilemma during surges, requiring automated decision-making to align with stakeholder aims.

2.2.5. Summary: Complementary and Conflicting Perspectives on AI Fairness Tensions

As summarized in Table 1, these four lenses illuminate distinct, complementary, and often conflicting aspects of AI fairness, revealing multilayered tensions. First, there is a fundamental conflict between process-based approaches (such as deontology) and outcome-based approaches (such as consequentialism). This tension is not only clear in the Ethics lens but is also embedded within Organizational Justice itself, which positions outcome-based (distributive) fairness to be balanced against process-based and interactional fairness in the context of an institution. Second, and just as critically, the purely outcome-based lenses are also in direct conflict. Economic Fairness prioritizes collective efficiency (Pareto optimality), which can be distribution-blind. Rawlsian Justice, in contrast, prioritizes redistributive equity for the least advantaged.

In sum, a “fair” solution from one perspective can be “unfair” from another. For example, an efficient allocation from Economic Fairness (maximizing total resources) could be deeply unfair from a Rawlsian perspective if it harms the worst off. No single lens is sufficient to understand the AI fairness problem, necessitating the synthesis across these perspectives to identify the common metatheoretical dimensions that cause these persistent conflicts.

Theory	Core Premise	Foci	Assumptions	Types of Fairness Tensions	Diagnosis of the AI Fairness Problem
Ethics (deontological & consequentialist)	Fairness is evaluated through competing ethical frameworks: deontology (duties, rights, procedural integrity) vs. consequentialism (aggregate outcomes/welfare)	Individual and societal	Moral priorities (duties vs. outcomes) evolve with societal norms and contexts	Incompatibilities between individual rights and collective benefits; dilemmas (e.g., removing sensitive attributes protects privacy but increases disparities)	Reveals fundamental incompatibilities (e.g., individual rights vs. collective utility) that create “no-win” dilemmas
Organizational Justice (distributive, procedural, interactional)	Fairness spans distributive equity (outcomes), procedural consistency and transparency (processes), and respectful, dignified treatment (interactions)	Individual and group	Institutional rules and stakeholder expectations are relatively stable.	Constraints and trade-offs—uniform transparency/regularity vs. context-sensitive, equity-oriented adjustments	Highlights the internal tensions of justice in an institutional context: procedural regularity (consistency), respectful treatment (interactional), and equitable outcomes (distributive)
Economic Fairness (collective efficiency vs. structural equity)	Fairness is framed by collective efficiency (Pareto optimality); redistribution is acceptable only if it does	Societal (macrolevel)	Well-functioning markets; structural inequities are externalities unless they	Trade-offs between collective efficiency and structural equity; efficiency-first designs	Highlights the trade-off between collective efficiency (which is distribution-blind) and

	not reduce aggregate welfare		depress aggregate welfare	can marginalize vulnerable groups	the risk of reinforcing structural inequities
Rawlsian Justice (equity prioritization for the least advantaged)	Fairness is anchored in the difference principle: inequalities are acceptable only if they improve the position of the least advantaged, consistent with equal basic liberties and fair equality of opportunity	Group and societal	Sufficient societal consensus and institutional capacity to implement redistributive policies and their constraints (e.g., equity weights, minimum guarantees)	Trade-offs between equity for disadvantaged groups and system efficiency that can harden into dilemmas when equity commitments strain capacity or predictive accuracy	Reveals a direct equity–efficiency conflict, where prioritizing the least advantaged can depress overall performance, especially under scarcity
Note: These four perspectives provide the theoretical foundations for FAIR’s theory of the problem. We synthesize them to identify three metatheoretical dimensions (principles, goals, foci) whose interdependencies within and among them are the root source of the AI fairness paradox.					

2.3. The Metatheoretical Dimensions of the AI Fairness Paradox

Building on this synthesis, we identify three metatheoretical dimensions of fairness tensions, which manifest as paradox within and among the dimensions:

- Principles*—incompatible definitions, constraints, and trade-offs across outcome-based, procedural, and interactional fairness.
- Goals*—trade-offs and dilemmas that pit fairness against model performance and business goals (e.g., efficiency, profitability).
- Foci*—dilemmas and trade-offs across individual, group, and societal levels.

This paradoxical nature reflects the interplay among organizational priorities, operational constraints, stakeholder values, and governance considerations (Fu et al., 2022; Kasy & Abebe, 2021; Kleinberg et al., 2018). Together, these dimensions and their interdependencies shape how fairness is operationalized in AI artifacts.

2.3.1. Principles Dimension: Incompatibilities, Constraints, and Trade-Offs

This dimension concerns how fairness ideals inform decision-making processes, governance, and stakeholder interactions. We categorize these tensions as follows:

- Outcome-based tensions* stem from incompatible assumptions about equity, equality, and need. For instance, equity-driven adjustments in healthcare AI can clash with equality policies, which mandate standardized treatments for all (Obermeyer et al., 2019).
- Procedural tensions* arise from endogenous (internal policy) and exogenous (regulatory) constraints and their trade-offs. For instance, transparency requirements in GDPR or HIPAA can conflict with privacy protections (Xu & Zhang, 2021). Meanwhile, endogenous process constraints (e.g., internally defined consistency rules) can conflict with equitable considerations, as in algorithmic hiring (Baird & Maruping, 2021).
- Interactional tensions* reflect standardization vs. personalization trade-offs (Robert et al., 2020). Standardization ensures consistency but may neglect individualized contexts, thus reducing perceived fairness, whereas personalization improves fit to the individual case but can undermine procedural consistency (Van den Broek et al., 2021).

Paradoxical Implications. These tensions show artifact design cannot rely on a single fairness principle. As societal expectations, regulations, and organizational contexts evolve, these tensions are guaranteed to resurface or intensify, amplifying the paradoxical nature of the AI fairness problem.

2.3.2. Goals Dimension: Trade-Offs Among Competing Objectives

These tensions arise from conflicts between fairness aims and organizational priorities, highlighting the complexities of integrating fairness in AI artifacts (Caton & Haas, 2024; Fu et al., 2022). Three critical trade-offs underlie them:

- Fairness vs. model performance* (accuracy, robustness, generalizability): Fairness constraints may reduce accuracy; conversely, optimizing for performance alone exacerbates bias (Cowgill & Tucker, 2020; Holm, 2023; Obermeyer et al., 2019).
- Fairness vs. operational efficiency*: Fairness-sensitive interventions increase computational complexity or latency, compromising efficiency or scalability (Fu et al., 2022; Teodorescu et al., 2021).

- *Fairness vs. economic performance*: In AI-automated decisions designed to enhance economic performance (e.g., targeted recommendations and ads), incorporating fairness objectives may lower engagement or profitability (Holm, 2023).

Paradoxical Implications. These trade-offs are dynamic rather than static. They demonstrate that fairness cannot be managed in a silo because a “fix” for one goal (e.g., fairness) will invariably impact others (e.g., cost or accuracy), making the resulting tensions inherently persistent and interdependent.

2.3.3. Foci Dimension: Dilemmas and Trade-Offs Across Levels.

Managing fairness at individual, group, and societal levels introduces characteristic dilemmas and trade-offs.

- *Resource-allocation dilemmas*: Maximizing societal fairness (e.g., population health via vaccine distribution) may disadvantage specific high-risk individuals who need prioritized access (Caton & Haas, 2024; Persad et al., 2009).
- *Cross-level trade-offs*: Expanding societal-level fairness (e.g., inclusive lending) can reduce individual-level precision, whereas strict individual-level fairness (e.g., precise diagnostics) can exacerbate societal disparities (Cowgill & Tucker, 2020; Teodorescu et al., 2021).

Paradoxical Implications. This dimension highlights the multilevel nature of the paradox. A decision that is fair at the societal level can be demonstrably unfair at the individual level, and vice versa. These different foci are in a constant, unavoidable state of tension.

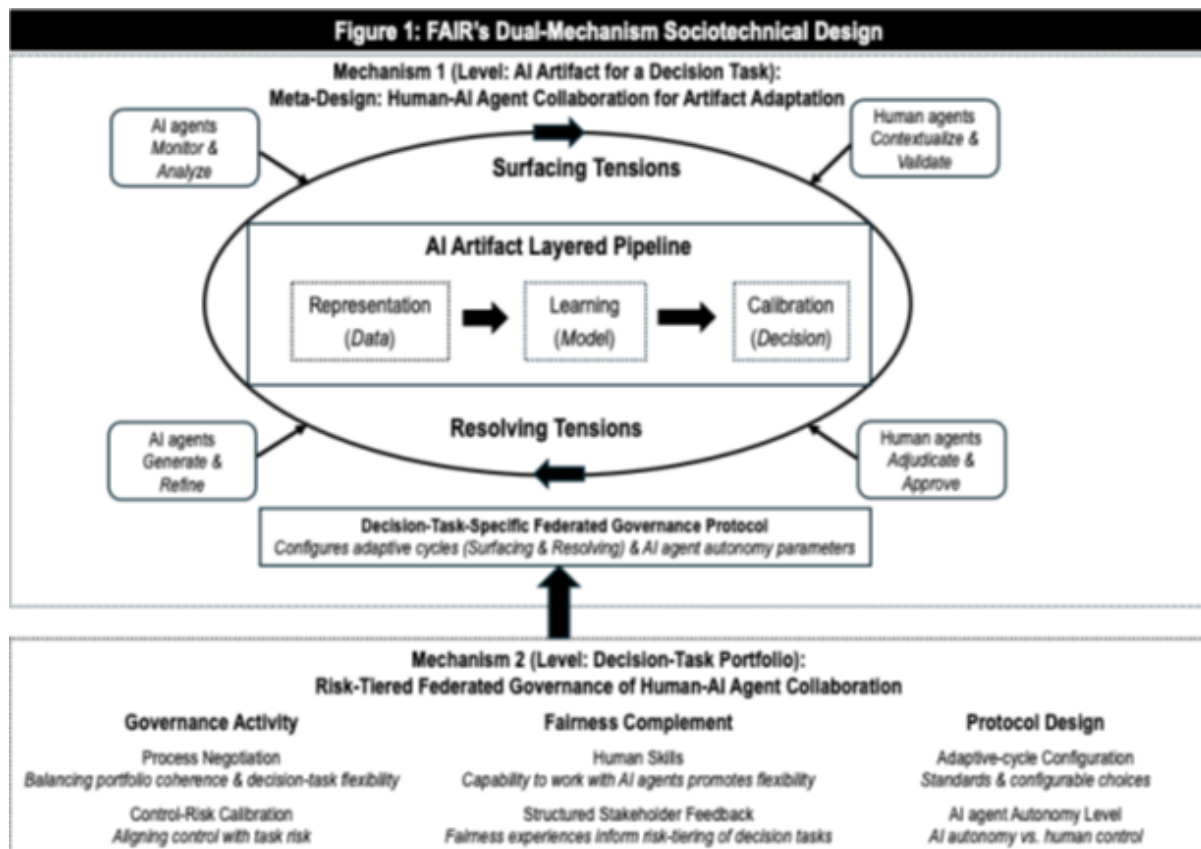
2.3.4. Summary: The Paradoxical Nature of the AI Fairness Problem.

Across the three metatheoretical dimensions of Principles, Goals, and Foci, fairness tensions manifest as constraints, incompatibilities, trade-offs, and dilemmas (summarized in Table 2). These tensions interact recursively, and shifting norms, strategies, and standards can reignite “settled” issues. Because of their persistent, interdependent, and dynamic nature, these tensions are paradoxical, meaning they cannot be resolved with one-off or static fixes.

Metatheoretical Dimension	Subdimensions	Theoretical Perspectives	Associated Tensions	Examples
Principles <i>How fairness is defined and operationalized</i>	Outcome-based, Procedural, Interactional	Ethics, Organizational Justice, Rawlsian Justice	Incompatibilities (equity vs. equality), constraints (transparency vs. privacy), trade-offs (standardization vs. personalization)	Equity interventions conflict with equality-based uniformity (Obermeyer et al., 2019); transparency mandates vs. privacy requirements (Xu & Zhang, 2021)
Goals <i>How fairness interacts with organizational priorities</i>	Fairness, Model performance, Economic/operational objectives	Economic Fairness, Organizational Justice	Trade-offs (fairness vs. accuracy/robustness, fairness vs. efficiency/scalability/cost, fairness vs. revenue/profitability)	Fairness constraints vs. accuracy in hiring (Holm, 2023); computational demands of fairness adjustments in real-time fraud detection (Fu et al., 2022)
Foci <i>At what level fairness is prioritized</i>	Individual, Group, Societal	Rawlsian Justice, Ethics, Organizational Justice	Dilemmas (group equity vs. individual rights), trade-offs (individual precision vs. societal inclusion)	Affirmative action disadvantaging qualified individuals (Caton & Haas, 2024); inclusive lending criteria reducing individual predictive precision (Cowgill & Tucker, 2020)
Summary <i>The Sociotechnical Paradox</i>	Interdependence within and among Principles, Goals, and Foci	Synthesis of perspectives	Continuous, recursive interplay of constraints, incompatibilities, trade-offs, and dilemmas arising from ongoing stakeholder contestation encoded within AI artifacts and continual influence from regulatory mandates and shifting societal norms, causing tensions to resurface dynamically	Equity-aware decision thresholds reshaping performance trade-offs; stakeholder contestation and regulatory shifts reactivating fairness tensions

3. The Theory of the Solution: FAIR's Dual-Mechanism Sociotechnical Design

To manage the paradox diagnosed earlier, FAIR advances a sociotechnical design with two reinforcing mechanisms (Figure 1). First, the core artifact-level mechanism (Mechanism 1) is informed by two sets of literature. The paradox lens (Raisch et al., 2018; Schad et al., 2016; Smith & Lewis, 2011) motivates the mechanism's responsive *process*: continuous adaptive cycles (Surfacing and Resolving). In parallel, IS design science (Abbasi et al., 2024; Hevner et al., 2004) and computer science (Russell & Norvig, 2009) inform its actors, distinguishing between AI artifacts (the “what,” which execute bounded decision tasks) and the human and AI agents (the “who”) responsible for adapting the artifacts. Second, the portfolio-level mechanism (Mechanism 2) is informed by the managing-with-AI literature (Asatiani et al., 2021; Grote et al., 2024; Hillebrand et al., 2025; Murray et al., 2021; Rai et al., 2019), which guides *how* to govern the persistent control–autonomy tension—which emerges from human–AI agent collaboration underlying Mechanism 1—across the portfolio of AI-automated decision tasks. We next elaborate these mechanisms in turn.



Note: Mechanism 2 (Portfolio-Level) establishes a governance structure that configures Mechanism 1 (Artifact-Level). Enabled by human skills and stakeholder feedback, central and local councils negotiate process configurations and calibrate AI agent autonomy levels to task risk. This resulting protocol structures the adaptive cycles (Surfacing and Resolving) executed by human and AI agents across the artifact layers.

3.1. Mechanism 1 (Artifact-Level): Human–AI Agent Collaboration for Artifact Adaptation

FAIR's first mechanism is the sociotechnical system that adapts a single AI artifact that automates a decision task. This mechanism has two core components: the process it follows (the adaptive cycles) and the actors

who perform it (the collaborating human and AI agents).

3.1.1. The Adaptive Cycles (Process)

This process operates through two iterative phases. First, the Surfacing phase entails the ongoing detection of tensions (e.g., incompatibilities, constraints, trade-offs, and dilemmas) arising from shifting regulatory demands, data drift, or evolving normative standards. In practice, the process continuously monitors the performance of AI-automated decisions, scanning for changes in data distributions or stakeholder expectations, and identifying emergent ethical concerns. By detecting subtle shifts, organizations can intervene early rather than reacting after harms escalate.

Second, Resolving centers on implementing targeted refinements—such as updating encodings or datasets (Representation), adjusting loss functions or parameters (Learning), or recalibrating decision policies (Calibration). These adjustments frequently require contextual knowledge to select the right interventions at the right time, thereby mitigating identified fairness risks while considering operational and economic objectives.

Because the mechanism applies to a modular, three-layer AI artifact architecture—Representation (data), Learning (model), and Calibration (decision) (Caton & Haas, 2024; Dolata et al., 2022; Mehrabi et al., 2021)—organizations can target fairness fixes at the point of emergence. By localizing issues and interventions, FAIR avoids costly, system-wide overhauls and instead executes targeted, incremental improvements (Abbasi et al., 2024). This layered, modular approach equips organizations to remain agile yet stable in meeting the continuous, evolving demands for fairness as conditions change. For example, if a virus variant increases false-negative rates for older patients in ED triage, Surfacing flags the divergence; Resolving updates age-related features and calibration thresholds and then re-audits subgroup error rates.

3.1.2. The Collaborative Actors (Who)

A distinguishing feature of FAIR is its clear separation of the AI artifact (the “what”) from the agents (the “who”) who execute the Surfacing and Resolving cycles:

- **AI artifact:** The trained, task-specific computational system—the modular Representation (data), Learning (model), and Calibration (decision) pipeline—that executes a bounded decision task (Abbasi et al., 2024; Hevner et al., 2004).
- **AI agents:** Computational actors that pursue assigned goals within guardrails (Russell & Norvig, 2009; Sutton & Barto, 2018). In FAIR, they execute the Surfacing (e.g., real-time detection) and Resolving (e.g., refinement generation) cycles to manage tensions.
- **Human agents:** Actors who supply contextual knowledge, ethical judgment, and policy oversight required for interpretation, prioritization, and authorization to manage the tensions.

This collaborative, sociotechnical mechanism creates synergy between the distinctive capabilities of AI and human agents and mitigates the failure modes of agent-only processes (Hillebrand et al., 2025; Rai et al., 2019). Counterfactually, an AI-only process lacks legitimacy, context, and ethical grounding, whereas a human-only process lacks the speed, scale, and continuous sensing needed to manage dynamic AI-automated decision systems.

For illustration, in ED triage, an AI agent detects that a recent intake shift has left a key demographic underrepresented in the training data and proposes synthetic augmentation and feature reweighting. A human agent (e.g., a clinical-policy board) reviews privacy and clinical plausibility, documents the rationale, and authorizes the change.

That said, these complementary strengths are neither static nor uniformly applied; nor can they be assumed to be without friction. How an organization orchestrates this collaboration depends on task risk and requires calibrating AI agent autonomy and human control.

3.2. Mechanism 2 (Portfolio-Level): Risk-Tiered Federated Governance

Human–AI agent collaboration creates a persistent control–autonomy tension: how much autonomy to grant AI agents versus how much oversight to retain by humans (Hillebrand et al., 2025; Murray et al., 2021; Robert et al., 2020). FAIR asserts this balance is dynamic and must be calibrated based on task risk and stakeholder feedback, beyond regulatory mandates.

Although this balance must be calibrated for each decision task, managing it in an ad hoc manner creates inconsistency and risk across the decision-task portfolio. First, to manage the portfolio, FAIR prescribes a federated structure. A central body sets global minimum process standards (e.g., approval thresholds, evidence requirements, escalation routes) to ensure global coherence. Meanwhile, local teams negotiate configurable choices to align the adaptive cycles with the context of a specific decision task, enabling local flexibility. This structure is enabled by the first fairness complement: human skills. Without the requisite skills to work with AI agents, central teams cannot

establish meaningful portfolio-wide standards and local teams cannot configure the collaboration protocol to leverage effective human-AI agent collaboration.

Second, within this federated structure, the level of AI agent autonomy is not uniform; it is risk-tiered. The risk profile of the decision task determines the “human–AI agent control-autonomy mix”—ranging from high AI autonomy (e.g., automatic execution with post hoc review) for low-risk tasks, to high human control (e.g., mandatory preapproval) for high-risk tasks. This policy is made responsive by the second fairness complement: structured stakeholder feedback. This feedback (e.g., from fairness dialogues) informs the risk classification, ensuring that AI agent autonomy is calibrated to community values and realities for the decision task.

Illustrating with ED triage, a central governance body in consultation with local clinical teams (Mechanism 2) may set a standard for routine drift bands on clinical criteria, while imposing tighter ICU escalation criteria thresholds in patients with certain comorbidities, accompanied by clinician oversight. The local clinical teams configure the ICU escalation workflows to follow this protocol, ensuring any proposed revisions to criteria thresholds are vetted by clinicians before deployment.

4. FAIR Requirements: Meta-requirements for AI Artifacts and Decision-Task Portfolio Governance

Building on the preceding diagnosis, we develop FAIR’s requirements (summarized in Table 3) for two focal targets: (i) the AI artifact that automates a decision task and (ii) the governance of the AI-automated decision task portfolio. For AI artifacts, we define MRs—the common properties that AI artifacts must support across the decision tasks they automate. We organize the artifact MRs into a three-layer computational pipeline for AI-automated decision-making—Representation (Data), Learning (Model), and Calibration (Decision)—and add a cross-layer MR that coordinates these layers within a decision task. This layering, supported by research on modularity (Baldwin & Clark, 2000; Yoo et al., 2010) and AI fairness (Caton & Haas, 2024; Mehrabi et al., 2021), enables separation of concerns, facilitates localized refinements without requiring substantial system-wide reconfigurations, and promotes continuous adaptation, while enhancing accountability and transparency in governance. Concurrently, to ensure coherence and flexibility across the portfolio of AI-automated decision tasks, we define portfolio-level GRs.

Rather than demanding a one-time resolution of tensions, the MRs specify what the artifact must support on an ongoing basis as priorities, regulation, and stakeholders’ expectations evolve (Caton & Haas, 2024; Mehrabi et al., 2021). Across layers, FAIR follows a simple progression—detect fairness tensions, balance fairness with other objectives, and assure coherence and accountability—which we instantiate as layer-specific MRs. We present these MRs by computational layer, specifying *what* fairness issues need to be surfaced and resolved. We reference these as MR–{layer} {index}, e.g., MR-R1, MR-L2, MR-C3, MR-CL1 and the portfolio-level GRs as GR–{index} (e.g., GR-1).

Scope (level)	Objective	Requirements
Artifact Level—Representation (Data) Layer	Encode fairness conditions in data by making definitions/policies explicit and correcting representational/quality distortions	MR-R1. Compute and expose fairness metrics (with versioned reports) to surface definitional incompatibilities MR-R2. Embed exogenous (regulatory, stakeholder) and endogenous (organizational) policies as explicit conditions that define boundaries for fairness representation MR-R3. Detect, flag, and remediate subgroup under-/overrepresentation and data-quality distortions
Artifact Level—Learning (Model) Layer	Structure fairness trade-offs by mitigating inherited distortions, balancing fairness with predictive performance, and restoring cross-level coherence	MR-L1: Implement model-level adjustments to mitigate distortions or incomplete data inherited from the Representation layer MR-L2: Compute and report fairness and predictive-performance indicators (including predictive uncertainty) to expose emerging trade-offs; tune objective-function and parameter constraints to achieve a workable balance MR-L3: Compute and report per-focus (individual, group, societal) error and fairness profiles to expose cross-level tensions; proportionally adjust cross-level interdependencies to restore balance

Artifact Level— Calibration (Decision) Layer	Resolve decision-level dilemmas by enabling transparent threshold/policy adjustment, balancing fairness with statistical and operational objectives, and assuring policy robustness	MR-C1: Embed calibration mechanisms to resolve fairness dilemmas through transparent decision-threshold and policy adjustments with rollback capability MR-C2: Recalibrate decision thresholds and policies to balance fairness with predictive performance and with operational/economic objectives MR-C3: Support policy simulation and counterfactual analysis, and per-group calibration reporting and stability tests
Artifact Level— Cross-Layer	Maintain artifact coherence by monitoring shifts across layers, containing error propagation at handoffs, and preserving audit-ready lineage and approvals	MR-CL1: Support cross-layer monitoring—with defined metrics, thresholds, and protocols—that surfaces emerging misalignments across the Representation, Learning, and Calibration layers over time MR-CL2: Mitigate error propagation across layers by enabling cross-layer impact assessment (pre/post change) and interface–contract verification at layer boundaries MR-CL3: Maintain a unified, immutable provenance and approval-and-authorization log for all fairness-relevant modifications to data (Representation), model (Learning), and decision policies (Calibration); support cross-layer traceability, versioning, and audit-ready reporting
Portfolio-Level— Decision-task Portfolio	Promote portfolio-level fairness coherence through federated governance and risk-tiered autonomy, providing a consistent and principled basis for configuring human–AI roles across decision tasks	GR-1: Define global minimum process standards for fairness evaluation, interpretability, and oversight, with explicit approval thresholds, evidence criteria, and escalation routes for surfacing and resolving fairness tensions across decision tasks GR-2: Adopt risk-tiered autonomy policies that specify when AI agents may act autonomously, when human preapproval is required, and when post hoc review suffices; update tiers based on reliability metrics and incident learning GR-3: Maintain structured stakeholder feedback channels (e.g., fairness dialogues) that inform central protocol updates and guide local adaptation, ensuring portfolio-level governance stays aligned with frontline realities
Note: MR codes refer to general properties of AI artifacts that automate decision tasks (regardless of specific models, datasets, or tasks). GR codes refer to organization-level GRs for the portfolio of AI-automated decision tasks.		

4.1. Representation (Data) Layer: Encoding Fairness Conditions

At the Representation layer, tensions arise from conflicting fairness definitions (e.g., equity vs. equality) and measures (Fu et al., 2022; Hurlin et al., 2026; Mitchell et al., 2021), exogenous or endogenous constraints (Cohen et al., 2022; Fu et al., 2022; Kordzadeh & Ghasemaghahi, 2022), and data-level issues such as subgroup under-/overrepresentation and data-quality distortions (Kane et al., 2021; Li et al., 2024; Rhue, 2023). To address these tensions, AI artifacts must encode and expose (i) the specific fairness definitions and metrics in use, (ii) relevant policy/constraint conditions, and (iii) subgroup representativeness and data quality—so human and AI agents can identify problematic areas and adjust data inputs.

First (MR-R1), to surface definitional incompatibilities, AI artifacts must compute and expose multi-metric, per-subgroup fairness assessments, because relying on a single metric (e.g., demographic parity) can obscure tensions between fairness principles (Binns, 2018; Mehrabi et al., 2021). Second (MR-R2), to define boundaries for fairness representation, AI artifacts must embed exogenous regulatory constraints (e.g., GDPR, anti-discrimination laws) and endogenous organizational policies (e.g., internal fairness guidelines) as explicit, versioned conditions that govern data representation and access (Cohen et al., 2022; Xu & Zhang, 2021). Third (MR-R3), to address representativeness and quality issues, the Representation layer must detect, flag, and remediate subgroup under-/overrepresentation and data-quality distortions. For example, hiring algorithms that fail to represent minority groups in training data can perpetuate demographic biases; expanding these datasets to include more diverse samples helps mitigate such risks (Fu et al., 2021; Obermeyer et al., 2019; Teodorescu et al., 2021).

4.2. Learning (Model) Layer: Structuring Fairness Trade-Offs

The Learning layer addresses trade-offs among fairness principles, goals, and foci. Beyond correcting data quality and representativeness at the Representation layer, compelling reasons exist to incorporate adjustments during learning. Simply balancing datasets (e.g., oversampling minority groups) can lead to overfitting the minority class or suppressing information from the majority class, whereas model-level techniques can flexibly handle imbalance without these drawbacks. Moreover, biases often emerge from feature interactions that may remain hidden during preprocessing, causing the model to perpetuate biased relationships (Kane et al., 2021); even

seemingly balanced data can encode implicit biases reflecting historical inequalities (Dolata et al., 2022; Kane et al., 2021).

First (MR-L1), to mitigate distortions inherited from the Representation layer, learning-layer interventions—such as representation learning, adversarial debiasing, and in-processing constraints that penalize biased outcomes and enforce fairness criteria—must reweight or suppress biased representations during training, thereby reducing the propagation of structural biases across layers (Hurlin et al., 2026; Kordzadeh & Ghasemaghahi, 2022; Teodorescu et al., 2021).

Second (MR-L2), to expose and balance fairness–performance trade-offs, artifacts must compute and report fairness and predictive-performance indicators—including predictive uncertainty—and tune objective-function and parameter constraints to achieve a workable balance. Third (MR-L3), to expose cross-level tensions and restore balance, artifacts must compute and report per-focus (individual, group, societal) error and fairness profiles and proportionally adjust cross-level interdependencies to manage individual–group–societal trade-offs. For example, embedding fairness-aware constraints within health-care AI models balances predictive accuracy with fairness, thereby mitigating systematic biases in underrepresented patient populations (Seyyed-Kalantari et al., 2021). Finally, making explicit how procedural transparency (e.g., GDPR compliance) interacts with outcome fairness ensures coherence across levels and surfaces violations requiring adjustment (Raghavan et al., 2020; Robert et al., 2020).

4.3. Calibration (Decision) Layer: Resolving Dilemmas and Trade-Offs

The Calibration layer translates model outputs into actionable decisions, serving as the critical point where predictions become consequences and unresolved tensions must be managed.

First (MR-C1), to resolve fairness dilemmas and trade-offs at the decision point, artifacts must embed transparent decision-threshold and policy-adjustment mechanisms with rollback capability. The artifacts should expose thresholds and allocation rules (e.g., cutoffs, weights/quotas, tie-breakers, capacity constraints, or priority queues that map scores to actions) and support safe, auditable policy change and rollback execution. Doing so enables local units to tackle cross-level tensions (individual–group–societal) explicitly and accountably.

Second (MR-C2), to balance fairness with statistical performance (e.g., accuracy, robustness) and with operational/economic objectives (e.g., latency/throughput, cost, risk limits), AI systems must recalibrate thresholds and policies governing decisions. For example, fairness-sensitive lending systems regularly tune approval cutoffs to widen financial inclusion while protecting profitability and portfolio risk (D'Amour et al., 2020; Jacobs & Wallach, 2021). This continuous balancing keeps decision policies and their linkages to model outputs aligned with shifting ethical expectations, operational constraints, and regulation.

Third (MR-C3), to make recalibration responsible and repeatable, AI artifacts must support policy simulation and counterfactual analysis, per-group calibration reporting, and stability tests. These capabilities allow teams to test *ex ante* adjustments, detect *ex post* drift, and document a calibration trail. Given intensifying scrutiny on unfair AI decisions (Teodorescu et al., 2021), proactive calibration reduces compliance risk, reputational harm (Rana et al., 2022), customer trust erosion (Zhdanov et al., 2022), and employee dissatisfaction (Rana et al., 2022).

4.4. Cross-Layer Coordination: Managing Tensions for Artifact Coherence

Although many fairness refinements focus on a single layer (e.g., data preprocessing at the Representation layer or in-processing constraints at the Learning layer), tensions often transcend these boundaries, evolving dynamically from data inputs to predictions and final calibration. This reflects the notion of *loose coupling* (Weick, 1976), where local interventions can yield unpredictable effects elsewhere; likewise, modular systems still require overarching coherence (Baldwin & Clark, 2000). Consequently, fairness issues can emerge or intensify if changes at one layer inadvertently introduce new biases at another.

First (MR-CL1), to manage cross-layer interdependencies over time, artifacts must support cross-layer monitoring—with defined metrics, thresholds, and protocols—that surfaces emergent misalignments across the Representation, Learning, and Calibration layers.

Second (MR-CL2), to prevent and contain error propagation, artifacts must support cross-layer impact assessment (pre-/post-change) and interface–contract verification at handoffs (schemas/encodings, units/scales/time windows, score/calibration status, dependency versions). This ensures each layer's outputs still satisfy the next layer's agreed inputs after any change.

Third (MR-CL3), to provide audit-ready lineage and approvals, artifacts must maintain a unified, immutable provenance and approval-and-authorization log for all fairness-relevant changes to data (Representation),

models (Learning), and decision policies (Calibration), with cross-layer traceability, versioning, and reporting. This creates a single source of truth for compliance and accountability.

4.5. Governance of the Decision-Task Portfolio

Managing with AI spans human-only, human–AI, and AI-only modes across tasks (Hillebrand et al., 2025; Murray et al., 2021). This complexity triggers persistent control–autonomy tensions—including opacity vs. interpretability, accountability vs. control, and over- vs. underreliance (Grote et al., 2024)—requiring robust, fairness-preserving governance to keep outcomes consistent across the decision-task portfolio.

Governance of AI-automated decision-making must address the organization’s decision-task portfolio—not only individual decision tasks. First (GR-1), to ensure portfolio coherence, the organization must define global minimum process standards for fairness evaluation, interpretability, and oversight, with approval thresholds, evidence criteria, and escalation routes for surfacing and resolving fairness tensions across decision tasks. Second (GR-2), to calibrate control–autonomy, the organization must adopt risk-tiered AI autonomy policies that specify when AI agents may act autonomously, when human preapproval is required, and when post hoc review suffices—updating tiers based on reliability metrics and incident learning. Third (GR-3), to sustain stakeholder alignment, the organization must maintain structured, inclusive stakeholder feedback channels (e.g., fairness dialogues) that inform protocol updates centrally and guide local adaptation—so portfolio governance remains aligned with frontline realities.

5. Operationalizing FAIR: Human–AI Agent Meta-Design and Portfolio-Level Governance

Building on the foregoing requirements—artifact MRs and portfolio GRs—in this section we propose (i) at the artifact level, a human–AI agent collaboration meta-design to adapt fairness within and across the Representation, Learning, and Calibration layers of the artifact pipeline, and (ii) at the AI-automated decision-task portfolio level, a federated governance mechanism with a risk-tiered AI agent autonomy policy. Rather than reactive, post hoc fixes, FAIR emphasizes continuous monitoring and preemptive refinement of artifacts that automate decision tasks through iterative Surfacing and Resolving cycles.

By clearly defining the roles, functions, and interactions of AI agents and human agents, the proposed meta-design aims to ensure adaptability, ethical robustness, and regulatory compliance (Fuster et al., 2022; Mehrabi et al., 2021). In the layer-specific subsections that follow, we detail how responsiveness operates at each computational layer, explaining how AI-agent-executed monitoring, prediction (e.g., early warning of drift or tension), and refinement—combined with human contextualization, ethical judgment, and oversight—address layer-specific tensions. To explicitly incorporate causality via counterfactual reasoning (Gregor, 2006), in each subsection we integrate a concise contrast (AI-only, human-only) to explain why the specified human-AI agent roles and interactions are superior for surfacing and resolving fairness tensions.

Following the layer-specific operationalization, we detail adaptive cycles for cross-layer coordination. We then operationalize portfolio governance—defining the role of global process standards and local configurable choices, risk-tiered AI agent autonomy (preapproval/automatic/post hoc), and stakeholder feedback—and show how these levers balance human control and AI agent autonomy across decision tasks while keeping fairness adaptation coherent and flexible.

We continue to use the ED-triage illustration to ground our operationalization. With this example and counterfactuals, we show how the artifact-level meta-design meets the MRs within and across layers, and how the portfolio-level governance design fulfills the GRs and dynamically calibrates the human–AI agent collaboration protocol to achieve fairness adaptation of AI artifacts across decision tasks.

5.1. Adaptive Cycles for the Representation (Data) Layer

Fairness adaptation at the Representation (Data) layer iteratively addresses data-level tensions through a human–AI agent collaboration mechanism. In Table 4, we summarize how each MR from the Representation-layer requirements (Table 3) is addressed in two complementary phases: Surfacing (AI agents: Monitor & Analyze; Human agents: Contextualize & Validate) and Resolving (AI agents: Generate & Refine; Human agents: Adjudicate & Approve)—consistent with FAIR’s responsiveness approach.

First (MR-R1), to surface definitional incompatibilities by computing and exposing per-subgroup fairness metrics, AI agents (Monitor & Analyze) continuously monitor multi-metric fairness by subgroup (e.g., demographic parity, equalized odds) to detect definitional conflicts (Kusner et al., 2017; Loftus et al., 2018). Human agents

(Contextualize & Validate) interpret those flags, assessing whether they indicate systemic distortions or permissible contextual variations, and guiding prioritization (Holstein et al., 2019).

After confirming the need for intervention, AI agents (Generate & Refine) apply methods such as synthetic data augmentation, subgroup reweighting, and fairness-aware encodings to propose targeted refinements aligned with relevant fairness definitions (Kasy & Abebe, 2021; Madras et al., 2018). Human agents (Adjudicate & Approve) finalize these refinements, ensuring definitional alignment and mitigating subgroup biases, particularly in high-stakes scenarios requiring explicit authorization (Selbst et al., 2019). Counterfactually, if AI agents alone handle Surfacing, definitional conflicts can be overemphasized, and if AI agents alone handle Resolving, adjustments may ignore domain meaning; by contrast, if human agents alone handle Surfacing, subtle metric divergences tend to go undetected, and if human agents alone handle Resolving, reconciliations risk being slow and weakly versioned.

Second (MR-R2), to define boundaries for fairness representations by embedding exogenous (regulatory, stakeholder) and endogenous (organizational) policies as explicit conditions, AI agents (Monitor & Analyze) identify compliance risks—such as violations of GDPR, HIPAA, or internal fairness policies—using methods such as privacy audits and attribute-leakage detection algorithms. Human agents (Contextualize & Validate) evaluate flagged issues against legal frameworks and organizational mandates, prioritize remediation, and set the remediation path.

AI agents (Generate & Refine) then apply compliance-oriented methods (e.g., differential privacy; attribute masking) for regulatory adherence (Dwork et al., 2012; Veale et al., 2018). Human agents (Adjudicate & Approve) authorize modifications, confirming audit-readiness and policy alignment (Raji et al., 2020; Selbst et al., 2019). Counterfactually, absent human agents during Surfacing, nonmaterial compliance variances are over-flagged; absent human agents during Resolving, privacy fixes can become unnecessarily restrictive, harming model performance. Conversely, absent AI agents during Surfacing, subtle risks like attribute leakage are missed; absent AI agents during Resolving, remediations are uneven and underdocumented.

Third (MR-R3), to detect, flag, and correct subgroup under- and overrepresentation and data-quality distortions, AI agents (Monitor & Analyze) surface distorted subgroup distributions, data-quality issues, and structural disparities affecting fairness, using methods such as distributional analysis, stratified missingness/quality profiling, and adversarial auditing (Kusner et al., 2017; Loftus et al., 2018). Human agents (Contextualize & Validate) differentiate genuine systemic biases from legitimate variations and prioritize actions (Holstein et al., 2019).

AI agents (Generate & Refine) then apply methods such as targeted data acquisition/augmentation, subgroup or propensity reweighting, and robust imputation to craft refined data representations and simulate effects on fairness and predictive performance (Kasy & Abebe, 2021). Human agents (Adjudicate & Approve) confirm that these adjustments correct identified biases without introducing new ones, and approve significant corrections (Suresh & Guttag, 2021). Counterfactually, with only human agents during Surfacing, latent disparities linger, and with only human agents during Resolving, fixes risk being slow and inconsistently scoped; with only AI agents during Surfacing, distributional idiosyncrasies trigger overcorrection, and with only AI agents during Resolving, new biases may be introduced.

Through this human–AI agent collaboration, the Representation layer establishes a robust data foundation for model learning and decision calibration. Consequently, we propose:

Design Principle 1 (Representation-Layer Fairness Adaptation)

(a) AI monitoring with human contextualization. AI agents continuously monitor data distributions and quality disparities using anomaly detection and multi-metric fairness assessments (e.g., demographic parity, predictive equality), escalating significant or ambiguous anomalies to human experts for deeper contextual validation—especially in high-stakes scenarios. Counterfactually, AI-only monitoring may over-flag issues without domain insight; human-only monitoring may delay or miss subtle problems.

(b) AI refinement with human adjudication. AI agents generate targeted data refinements (e.g., synthetic data augmentation, subgroup reweighting), which human experts and compliance specialists then evaluate for final approval—ensuring contextual appropriateness, ethical justification, and legal compliance. Counterfactually, AI-only refinement may embed subtle biases, whereas human-only refinement may be slow or partial. Iterative collaboration provides both analytical precision and contextual awareness.

Illustrative example—ED triage (Representation layer). Patient records from underserved communities may be incomplete or erroneous, risking underrepresentation. AI agents (Monitor & Analyze) detect anomalous subgroup distributions or missing attributes, prompting human agents (Contextualize & Validate) to confirm whether these data gaps reflect structural biases or legitimate variations. If potential biases are confirmed, AI agents (Generate & Refine) apply methods such as synthetic data augmentation or reweighted features to better represent at-risk

demographics. Human agents (Adjudicate & Approve) ensure alignment with privacy regulations (e.g., HIPAA) and clinical standards, authorizing changes only after a thorough ethical and operational review. A responsive collaboration at the Representation layer best balances timely detection with contextual judgment.

Although data-level refinements mitigate foundational biases, they are not a complete solution. New fairness tensions can emerge during model learning and decision calibration, which is why FAIR's modular, iterative approach must continue at subsequent layers.

Table 4. Operationalizing Fairness Adaptation at the Representation (Data) Layer

Meta-Requirements	Surfacing Phase		Resolving Phase	
	AI Agents (Monitor & Analyze)	Human Agents (Contextualize & Validate)	AI Agents (Generate & Refine)	Human Agents (Adjudicate & Approve)
MR-R1. Compute and expose fairness metrics (with versioned reports) to surface definitional incompatibilities	Compute and expose multi-metric fairness indicators at dataset and feature levels; produce versioned reports highlighting definitional conflicts (e.g., equity vs. equality) Illustrative methods: multi-metric fairness indicators (e.g., demographic parity; equal opportunity; equalized odds); versioned metric reports; multi-metric fairness audits	Interpret metric conflicts in light of policy/ethical definitions and stakeholder expectations; determine whether conflicts require data adjustments or documentation with rationale	Prepare metric-consistent data adjustments and reporting views to reconcile definitional choices (naming, semantics); align metric definitions across teams Illustrative methods: metric-constrained reweighting; fairness-aware encodings; report/documentation generators	Approve metric definitions and adjustments; confirm clarity and versioning, and verify that reconciliations mitigate subgroup biases in high-stakes settings
MR-R2. Embed exogenous (regulatory, stakeholder) and endogenous (organizational) policies as explicit conditions that define boundaries for fairness representation	Scan data/feature schemas for privacy, anti-discrimination, and policy compliance issues (e.g., GDPR, HIPAA; internal fairness rules) Illustrative methods: privacy audits; attribute-leakage checks; policy-as-code validators	Validate the mapping of policies to data conditions (sensitive attributes, access scopes, retention); assess materiality and determine remediation path for compliance issues	Encode policy conditions into schemas, access controls, and lineage metadata; produce evidence for audit Illustrative methods: schema annotations; access-control lists; lineage links; differential-privacy or privacy-enhancing transformations	Approve compliance-oriented refinements and any exceptions; confirm audit readiness, stakeholder alignment, and policy conformance
MR-R3. Detect, flag, and remediate subgroup under-/overrepresentation and data-quality distortions	Monitor subgroup coverage, label quality, and data integrity; flag under-/overrepresentation and distortions that affect fairness Illustrative methods: distributional analysis; missingness/quality profiling; stratified label-noise checks; adversarial auditing	Assess materiality and potential harm of detected disparities; prioritize remediation scope and sequencing (e.g., resampling, acquisition, relabeling)	Apply representativeness/quality fixes and simulate effects on fairness and predictive performance. Illustrative methods: targeted acquisition/augmentation; subgroup or propensity reweighting; debiasing encodings; robust imputation; post-remediation fairness audits	Approve corrections; verify restored representativeness/quality and absence of new biases; confirm documentation and audit readiness
Note: In routine or low-stakes settings, phases often follow a linear flow—AI agents (Monitor & Analyze) → Human agents (Contextualize & Validate) → AI agents (Generate & Refine) → Human agents (Adjudicate & Approve)—with minimal iteration. In complex or high-stakes settings, phases iterate with deeper human contextualization, rigorous ethical scrutiny, and repeated human–AI agent interactions. All approvals are recorded in the unified provenance and approval-and-authorization log.				

5.2. Adaptive Cycles for the Learning (Model) Layer

Fairness adaptation at the Learning (Model) layer iteratively addresses model-level fairness tensions through a human–AI agent collaboration mechanism. In Table 5, we summarize how each MR from the Learning-layer requirements (Table 3) is met in two complementary phases: Surfacing (AI agents: Monitor & Analyze, Human agents: Contextualize & Validate) and Resolving (AI agents: Generate & Refine, Human agents: Adjudicate & Approve)—consistent with FAIR's responsiveness approach.

First (MR-L1), to implement model-level adjustments to mitigate distortions inherited from the Representation layer, AI agents (Monitor & Analyze) track and detect fairness tensions emerging from inherited data issues. Methods include sensitivity analyses, model-drift checks, and fairness-aware loss-function audits to surface biases resulting from incomplete or distorted data representations (Benbya et al., 2020; Zafar et al., 2017). Human agents (Contextualize & Validate) judge materiality—differentiating systemic distortion from acceptable contextual variation—and set remediation priority (Holstein et al., 2019; Selbst et al., 2019).

Once validated, AI agents (Generate & Refine) propose targeted refinements—using methods such as adversarial debiasing or loss-function recalibration—aimed at mitigating inherited data distortions without undermining interpretability or compliance (Agarwal et al., 2018; Madras et al., 2018). Human agents (Adjudicate & Approve) finalize these adjustments, ensuring that proposed changes address data-driven biases while meeting ethical and operational standards (Veale et al., 2018; Wachter & Mittelstadt, 2019). If only AI agents handle Surfacing, minor signals can get overprioritized, and if only AI agents handle Resolving, parameter changes may undermine interpretability or compliance; whereas with only human agents in Surfacing, subtle distortions may go undetected, and with only human agents in Resolving, fixes may lag and lack consistent documentation.

Second (MR-L2), to compute and report fairness and predictive-performance indicators (including predictive uncertainty) to expose emerging trade-offs, and to tune objective-function and parameter constraints to achieve a workable balance, AI agents (Monitor & Analyze) continually track fairness metrics alongside predictive performance and uncertainty, flagging emerging fairness-performance tensions, such as equity versus accuracy or robustness (Mehrabi et al., 2021; Mitchell et al., 2021). Human agents (Contextualize & Validate) interpret the flagged trade-offs in light of strategy, feasibility, and stakeholder expectations to set escalation thresholds (Fuster et al., 2022; Wang & Siau, 2019).

When escalation is warranted, AI agents (Generate & Refine) adjust model parameters—such as revising loss functions or constraint weights and running multi-objective optimization—documenting these revisions for human evaluation (Madras et al., 2018). Human agents (Adjudicate & Approve) confirm these refinements, ensuring alignment with strategy, technical viability, ethical guidelines, and regulatory mandates (Selbst et al., 2019). AI-only Surfacing risks over-adjusting to trade-off noise, whereas human-only Surfacing misses nuanced conflicts; in Resolving, AI-only tweaks risk model brittleness, whereas human-only balancing is slow, ad hoc, and thinly documented.

Third (MR-L3), to compute and report per-focus (individual, group, societal) error and fairness profiles to expose cross-level tensions and proportionally adjust interdependencies to restore balance, AI agents (Monitor & Analyze) assess model outputs to detect cross-level fairness tensions—for example, when interventions that enhance group-level diversity erode individual fairness (Benbya et al., 2020; Chouldechova, 2017). Human agents (Contextualize & Validate) interpret these tensions using context, ethical guidelines, and compliance frameworks to determine which trade-offs require resolution (Raji et al., 2020; Veale et al., 2018).

After confirming that cross-level fairness tensions warrant remediation, AI agents (Generate & Refine) propose strategies—using methods such as fairness-aware optimization or counterfactual algorithms—and document each proposal (Agarwal et al., 2018; Kusner et al., 2017). Human agents (Adjudicate & Approve) evaluate and finalize these refinements, striving to balance fairness goals with other objectives. Counterfactually, AI-only Surfacing tends to prioritize what metrics capture and underweight harms that are unmeasured or imperfectly proxied (e.g., norm- and context-dependent harms), and AI-only Resolving can overfit formal constraints; by contrast, human-only Surfacing misses weak cross-level signals, and human-only Resolving is slow and thinly documented.

Thus, the AI–human agent collaboration at the Learning layer surfaces and resolves fairness tensions, maintaining alignment with ethical, contextual, and operational considerations (Holm, 2023; Jarzabkowski et al., 2013). Accordingly, we propose the following design principle:

Design Principle 2 (Learning-Layer Fairness Adaptation)

(a) AI monitoring with human contextualization. AI agents continuously monitor model behaviors—for example, by performing sensitivity analyses, fairness-aware loss-function audits, or cross-level fairness checks—and escalate significant distortions, trade-offs, or anomalies to human agents for interpretation and prioritization, particularly in high-stakes scenarios that require iterative human–AI agent interaction. Counterfactually, AI-only monitoring may over-flag issues without domain insight, whereas human-only monitoring may delay or miss subtle problems.

(b) AI refinement with human adjudication. AI agents adjust model parameters and manage cross-level fairness interdependencies using validated methods (e.g., adversarial debiasing, counterfactual fairness, constraint reweighting, multi-objective optimization). Human agents adjudicate these proposed refinements, ensuring ethical appropriateness, contextual coherence, and operational feasibility—especially for high-stakes interventions. Counterfactually, AI-only refinements may be rigid or ethically naïve, whereas human-only refinements may be

slow or partial. Balancing AI's systematic adjustments with human oversight delivers both efficiency and contextual rigor.

Illustrative example—ED triage (Learning layer): Even after data-oriented refinements (the Representation-layer cycle described earlier), fairness tensions may persist in predictive triage models. AI agents (Monitor & Analyze) detect ongoing group disparities or emergent fairness-performance trade-offs, prompting human agents (Contextualize & Validate) to determine whether these issues justify model adjustments. AI agents (Generate & Refine) then propose parameter modifications—such as adjusting loss functions to better balance equitable treatment with clinical accuracy. Human agents (Adjudicate & Approve) assess and finalize these refinements, ensuring they address fairness-performance tensions without introducing unintended harm. A human–AI agent collaboration mechanism reconciles fairness and predictive-performance objectives in high-stakes ED triage while maintaining clinical safety and compliance.

Table 5. Operationalizing Fairness Adaptation at the Learning (Model) Layer

Meta-Requirements	Surfacing Phase		Resolving Phase	
	AI Agents (Monitor & Analyze)	Human Agents (Contextualize & Validate)	AI Agents (Generate & Refine)	Human Agents (Adjudicate & Approve)
MR-L1. Implement model-level adjustments to mitigate distortions or incomplete data inherited from the Representation layer	Monitor model behavior/predictions for biases caused by inherited data distortions Illustrative methods: sensitivity analyses; model-drift checks; fairness-aware loss audits	Judge materiality—systemic distortion vs. acceptable variation; set remediation priority/scope for high-stakes cases	Adjust parameters/algorithms to correct inherited distortions Illustrative methods: fairness-constrained optimization; adversarial debiasing; loss-function recalibration	Approve refinements that mitigate distortion without creating new biases; verify interpretability and compliance requirements
MR-L2. Compute and report fairness and predictive-performance indicators (including predictive uncertainty) to expose emerging trade-offs; tune objective-function and parameter constraints to achieve a workable balance	Track fairness and predictive-performance indicators and compute predictive uncertainty; flag emerging fairness–performance trade-offs Illustrative methods: fairness–performance dashboards and reports; trade-off sensitivity analysis; probability calibration tests	Interpret flagged trade-offs in light of strategy, feasibility, and stakeholder expectations; set escalation criteria for material conflicts	Tune constraints/weights to reach a workable balance Illustrative methods: constrained optimization; fairness-aware parameter tuning	Evaluate and approve revised trade-offs; verify operational viability, policy fit, and ethical acceptability
MR-L3. Compute and report per-focus (individual, group, societal) error and fairness profiles to expose cross-level tensions; proportionally adjust interdependencies to restore balance	Compute/report per-focus error and fairness profiles; flag individual–group–societal tensions Illustrative methods: multilevel fairness audits; subgroup sensitivity analyses	Determine rebalancing across levels; set scope and sequencing for cross-level adjustments	Recalibrate interdependencies (e.g., subgroup weighting; hierarchical/multilevel modeling) to restore cross-level balance Illustrative methods: multilevel modeling techniques; subgroup reweighting algorithms	Confirm and approve adjustments that preserve cross-level coherence without ethical or operational regressions
Note: In routine or low-stakes settings, phases typically follow a linear sequence—AI agents (Monitor & Analyze) → Human agents (Contextualize & Validate) → AI agents (Generate & Refine) → Human agents (Adjudicate & Approve)—with minimal iteration. In complex or high-stakes settings, phases iterate with deeper human contextualization, rigorous ethical scrutiny, and repeated human–AI agent interactions. All approvals are recorded in the unified provenance and approval-and-authorization log.				

5.3. Adaptive Cycles for the Calibration (Decision) Layer

Fairness adaptation at the Calibration (Decision) layer iteratively addresses decision-policy tensions through a human–AI agent collaboration mechanism. In Table 6, we summarize how each MR from the Calibration-layer requirements (Table 3) is met in two complementary phases—Surfacing (AI agents: Monitor & Analyze; Human agents: Contextualize & Validate) and Resolving (AI agents: Generate & Refine; Human agents: Adjudicate & Approve)—consistent with FAIR's responsiveness approach.

First (MR-C1), to embed calibration mechanisms that resolve fairness dilemmas through transparent decision-threshold and policy adjustments (with rollback capability), AI agents (Monitor & Analyze) continuously assess thresholds and policies for dilemmas using methods such as confidence-interval calibration and subgroup outcome-disparity tests (Chouldechova, 2017; D'Amour et al., 2020). Human agents (Contextualize & Validate), guided by ethical frameworks, regulatory mandates, and organizational policies, determine whether flagged threshold instabilities reflect systemic ethical concerns or permissible trade-offs and set escalation criteria and scope (Raji et al., 2020; Selbst et al., 2019).

Once validated, AI agents (Generate & Refine) propose fairness-constrained recalibration and robust thresholding for evaluation (Holstein et al., 2019; Madras et al., 2018). Human agents (Adjudicate & Approve) authorize changes that are ethically justifiable, operationally sound, and legally compliant (Fuster et al., 2022; Wachter & Mittelstadt, 2019). If only AI agents handle Surfacing, borderline threshold shifts can be treated as material issues, and if only AI agents handle Resolving, recalibrations may sidestep operational and ethical constraints; whereas with only human agents in Surfacing, subtle instabilities tend to go unnoticed, and with only human agents in Resolving, changes are slow, uneven, and thinly documented.

Second (MR-C2), to recalibrate decision thresholds and policies to balance fairness with predictive performance and with operational/economic objectives, AI agents (Monitor & Analyze) track emerging fairness–predictive-performance tensions and fairness–operational/economic tensions (e.g., latency/throughput, cost, risk limits), including model-induced instabilities affecting decision consistency (Green & Chen, 2019; Veale et al., 2018). Human agents (Contextualize & Validate), drawing on domain and compliance knowledge, judge acceptability and priority of these tensions (Benbya et al., 2020; Wang & Siau, 2019).

Subsequently, AI agents (Generate & Refine) propose threshold and policy refinements using methods—e.g., fairness-, cost-, and risk-aware threshold-robustness optimization and probability calibration—and document proposed changes (Holstein et al., 2019; Mitchell et al., 2021). Human agents (Adjudicate & Approve) ratify these recalibrated thresholds, ensuring they manage cumulative errors without introducing biases or degrading performance (Selbst et al., 2019). If only AI agents handle Surfacing, subtle accumulations risk being overcorrected as if universally material, and if only AI agents handle Resolving, adjustments can sidestep operational and ethical constraints; whereas with only human agents handling Surfacing, faint drifts and compounding errors tend to be under-detected, and with only human agents handling Resolving, responses are reactive, piecemeal, and thinly documented.

Third (MR-C3), to support policy simulation and counterfactual analysis, and per-group calibration reporting and stability tests, AI agents (Monitor & Analyze) assess the stability and subgroup calibration of thresholds and policies, flagging drift or instability using methods like per-group calibration curves and subgroup stability tests. Human agents (Contextualize & Validate) judge materiality and decide whether to evaluate simulated alternatives.

After confirmation, AI agents (Generate & Refine) run policy simulations and counterfactual analyses to compare candidate threshold and policy changes and their subgroup impacts; human agents (Adjudicate & Approve) select and approve the change that meets fairness and predictive-performance objectives with acceptable operational risk. Counterfactually, if only AI agents execute Surfacing, transient fluctuations risk being treated as material instability; if only AI agents execute Resolving, simulations can chase metric gains while overlooking operational constraints. By contrast, with only human agents in Surfacing, subtle subgroup miscalibration tends to go undetected; with only human agents in Resolving, scenario coverage may be narrow, and documentation can be thin.

Through this human–AI agent collaboration, the Calibration layer's decision policies remain aligned with evolving ethical standards, operational objectives, and regulatory mandates. Accordingly, we propose:

Design Principle 3 (Calibration-Layer Fairness Adaptation)

(a) AI monitoring with human contextualization. AI agents continuously monitor decision thresholds and policies (e.g., threshold-sensitivity analysis; subgroup-specific disparity checks; probability calibration tests), surfacing potential fairness dilemmas and fairness-performance trade-offs; human agents perform ethical evaluation, contextual interpretation, and compliance review, especially in high-stakes scenarios. Counterfactually, AI-only monitoring may elevate metric signals without sufficient operational or ethical context, whereas human-only monitoring may miss subtle or accumulating calibration anomalies.

(b) AI refinement with human adjudication. AI agents generate fairness-sensitive adjustments to decision thresholds and policies (e.g., fairness-constrained recalibration; robust thresholding; cost-/risk-aware optimization). In high-stakes contexts, these refinements require iterative human adjudication to ensure proposed changes remain ethically justified, operationally feasible, and compliant with regulatory standards. Counterfactually, AI-only refinement may sidestep domain and governance constraints, whereas human-only refinement tends to be slower, piecemeal, and thinly documented.

Illustrative example—ED triage (Calibration layer). Even after addressing data and model biases, fairness tensions can persist when calibrating patient triage thresholds. Suppose tightening thresholds to reduce false positives disproportionately increases false negatives for a vulnerable demographic. AI agents (Monitor & Analyze) detect disparities and quantify the shift, prompting human agents (Contextualize & Validate) to determine whether the resulting trade-offs are ethically defensible or operationally permissible. If tensions warrant intervention, AI agents (Generate & Refine) propose fairness- and risk-aware recalibration using methods like constraint reweighting or robust thresholding to simulate subgroup impacts. Human agents (Adjudicate & Approve) confirm that these adjustments conform to clinical, ethical, and organizational mandates, and approve implementation with rollback parameters and post-deployment monitoring. This human–AI mechanism at the Calibration layer preserves fairness-performance balance while maintaining clinical safety and compliance.

Although these layer-specific mechanisms are essential for addressing localized tensions, they do not operate in isolation. Refinements at one layer can propagate biases or create new trade-offs at others, necessitating the cross-layer coordination mechanism detailed next.

Table 6. Operationalizing Fairness Adaptation at the Calibration (Decision) Layer

Meta-Requirements	Surfacing Phase		Resolving Phase	
	AI Agents (Monitor & Analyze)	Human Agents (Contextualize & Validate)	AI Agents (Generate & Refine)	Human Agents (Adjudicate & Approve)
MR-C1. Embed calibration mechanisms to resolve fairness dilemmas through transparent decision-threshold and policy adjustments (with rollback capability)	Continuously assess decision thresholds and policies for fairness dilemmas and subgroup-specific impacts Illustrative methods: threshold-sensitivity analysis; subgroup outcome-disparity checks	Evaluate whether flagged dilemmas reflect biases or acceptable trade-offs; define escalation criteria, scope, and decision criteria for high-stakes cases	Generate fairness-sensitive threshold/policy adjustments to reconcile dilemmas; package with rollback parameters Illustrative methods: fairness-constrained recalibration; robust thresholding	Authorize recalibration; confirm integrity, feasibility, and compliance; verify rollback readiness
MR-C2. Recalibrate decision thresholds and policies to balance fairness with predictive performance and with operational/economic objectives	Monitor emerging fairness–predictive-performance and fairness–operational/economic trade-offs (e.g., latency/throughput, cost, risk limits) and model-induced instabilities affecting decision consistency Illustrative methods: trade-off analytics; latency/throughput and cost/risk monitors	Judge acceptability and priority; apply ethical, operational, and regulatory guidance to set scope and sequencing	Refine decision thresholds and policies to address identified trade-offs and any propagated error Illustrative methods: threshold-robustness optimization; fairness-aware and cost-aware adjustments	Approve final refinements; verify alignment with fairness commitments, predictive-performance standards, and operational/economic constraints, and regulatory requirements
MR-C3. Support policy simulation and counterfactual analysis, and per-group calibration reporting and stability tests	Assess policy stability and subgroup calibration; flag drift/instability Illustrative methods: per-group calibration curves; subgroup stability tests	Determine whether simulated alternatives are warranted; set scope and decision criteria for adoption	Run policy simulations and counterfactual analyses; compare candidate policies and subgroup impacts Illustrative methods: counterfactual evaluation; scenario analysis	Select and approve the preferred policy/threshold change; register post-change monitoring thresholds and acceptance criteria
Note: In routine or low-stakes settings, phases typically follow a linear sequence—AI agents (Monitor & Analyze) → Human agents (Contextualize & Validate) → AI agents (Generate & Refine) → Human agents (Adjudicate & Approve)—with minimal iteration. In complex or high-stakes settings, phases iterate with deeper contextualization, rigorous ethical scrutiny, and repeated human–AI agent interactions. All approvals are recorded in the unified provenance and approval-and-authorization log.				

5.4. Adaptive Cycles for Cross-Layer Coordination

Fairness adaptation can falter if interventions at one layer (Representation, Learning, or Calibration) do not account for downstream impacts or upstream constraints in the AI artifact pipeline. Cross-layer coordination ensures that each layer’s refinements are coherent and aligned with ethical, operational, and regulatory objectives (Murray et al., 2021; Teodorescu et al., 2021). By considering the interdependencies across the pipeline, cross-layer

coordination provides consistent oversight, preventing myopic fixes at one layer that might inadvertently undermine system-wide fairness or performance. In Table 7, we summarize how each cross-layer MR from the Cross-layer requirements (Table 3) is met through the human–AI agent collaboration mechanism.

First (MR-CL1), to manage cross-layer interdependencies, artifacts must support monitoring—with metrics, thresholds, and protocols—that surface misalignments across the Representation, Learning, and Calibration layers. AI agents proactively monitor fairness signals (e.g., drift, subgroup outcome disparities, instability) at each layer, detect cross-layer trends, and trigger impact checks, whereas human agents interpret escalations, determine cadence and scope, and route high- vs. low-stakes issues appropriately (Abbasi et al., 2024; Fuster et al., 2022).

Once an intervention is approved, AI agents (Generate & Refine) execute the fairness (re)calibration across the affected layers. Human agents (Adjudicate & Approve) then review and approve the final cross-layer intervention, confirming that the scope is proportionate and that all stakeholder and regulatory sign-offs are complete. Counterfactually, if only AI agents handle Surfacing, ambiguous fluctuations may be over-escalated as material, and if only AI agents handle Resolving, follow-on actions may ignore organizational priorities; whereas with only human agents in Surfacing, subtle cross-layer shifts tend to be under-detected, and with only human agents in Resolving, responses are slow, uneven, and thinly documented.

Second (MR-CL2), to mitigate error propagation across layers, artifacts must enable cross-layer impact assessment (pre/post change) and interface–contract verification at layer handoffs. AI agents (Monitor & Analyze) trace anomalies from the Calibration (Decision) layer back to the upstream model or data causes via automated auditing (Benbya et al., 2021; Chouldechova, 2017), and human agents (Contextualize & Validate) decide where coordinated multilayer interventions are warranted and in what sequence (Fu et al., 2022). Without end-to-end checks, local fixes can exacerbate errors; these controls align adjustments with goals, standards, and values (Murray et al., 2021; Wachter & Mittelstadt, 2019).

AI agents (Generate & Refine) then produce a remediation plan, such as staged data, model, or policy updates, and rerun impact checks and acceptance tests. Human agents (Adjudicate & Approve) approve the final remediation and its sequencing, verifying that all post-change acceptance tests passed, no unresolved interface breaks remain, and rollback readiness is documented. Counterfactually, if only AI agents handle Surfacing, interface breaks may be flagged without context, and pre/post comparisons can overweight metric noise; if only AI agents handle Resolving, fixes risk being mis-sequenced across layers. By contrast, if only human agents handle Surfacing, quiet interface drift tends to be missed, and if only human agents handle Resolving, coordination lags and dependencies are overlooked.

Third (MR-CL3), to maintain a unified, immutable provenance and approval-and-authorization log for all fairness-relevant modifications to data (Representation), model (Learning), and decision policies (Calibration)—and to support cross-layer traceability, versioning, and audit-ready reporting—artifacts must provide a single source of truth linking who/what/when/why to evidence (e.g., simulations and impact assessments). For this unified provenance system, AI agents record fairness interventions and their rationales (Benbya et al., 2020), whereas human agents verify completeness and policy compliance. This pipeline-wide system fosters accountability by tracing anomalies, facilitating compliance, and establishing rollback readiness without duplicate logging in each layer (Jacobs & Wallach, 2021).

AI agents (Generate & Refine) integrate this logging instrumentation into the continuous deployment pipeline, enforce version linkage, and generate audit-ready reports. Human agents (Adjudicate & Approve) certify the final audit readiness and compliance, confirming that the evidence is sufficient for independent review. Counterfactually, absent human verification during Surfacing, entries may be incomplete or weakly justified, and absent AI support during Resolving, lineage updates tend to lag and cross-references are error-prone; a unified, audit-ready lineage maintained by AI and human agents mitigates both risks.

In synthesis, cross-layer coordination aligns each layer’s adaptive cycles with organizational triggers, ensuring that interventions remain consistent with evolving strategies, capabilities, and constraints (Fuster et al., 2022; Hillebrand et al., 2025). Rather than confining Surfacing and Resolving loops to each layer in isolation, cross-layer coordination uses human–AI agent collaboration to determine necessary refinements—whether at Representation, Learning, or Calibration. By clarifying escalation paths and enabling feedback loops, organizations continuously calibrate fairness, preventing localized fixes from undermining system-wide goals. In doing so, cross-layer coordination supports an integrated, adaptive approach, aligning fairness with ethical and strategic commitments. Accordingly, we advance:

Design Principle 4 (Cross-Layer Fairness Coordination)

(a) AI monitoring with human contextualization. AI agents continuously track fairness signals across the Representation, Learning, and Calibration layers and run cross-layer impact assessment (pre/post change) with interface–contract verification; human agents interpret anomalies, set escalation criteria and cadence, authorize

interventions, and finalize coordinated refinements—especially in high-stakes scenarios. Counterfactually, AI-only monitoring may over-escalate ambiguous anomalies or miss interface breaks; human-only monitoring may under-detect subtle cross-layer shifts or propagate local fixes without end-to-end checks.

(b) AI refinement with human adjudication. AI agents generate coordinated cross-layer refinements (e.g., staged data/model/policy updates), implement pre/post impact checks and interface–contract verification, and orchestrate rollout/backout; human agents adjudicate and approve the integrated plan, verifying ethical/operational alignment and regulatory compliance. Counterfactually, AI-only refinement can mis-sequence changes or miss governance constraints; human-only refinement can be slow, fragmented, or incomplete.

Illustrative example—ED triage (Cross-layer coordination). Assume a hospital has refined data representation and model learning to mitigate biases, yet new tensions emerge after calibrating the triage thresholds for each demographic group. AI agents track how triage policies yield unanticipated equity gaps among subgroups, linking effects to subtle model overshoots. Human agents then evaluate whether these cross-layer disparities require adjusting thresholds or returning to model-level constraints, potentially reweighting certain features. By maintaining the unified provenance and approval-and-authorization log of data, model, and threshold changes, governance committees can holistically assess each proposed fix’s implications. Had each layer acted in isolation, data or model refinements might have skewed triage decisions for certain at-risk subgroups. Instead, cross-layer coordination ensures that administrators, compliance officers, and domain experts collectively determine which interventions remain consistent with patient safety standards, ethical guidelines, and privacy regulations.

Table 7. Operationalizing Fairness Adaptation for Cross-Layer Coordination

Meta-Requirements	Surfacing Phase		Resolving Phase	
	AI Agents (Monitor & Analyze)	Human Agents (Contextualize & Validate)	AI Agents (Generate & Refine)	Human Agents (Adjudicate & Approve)
MR-CL1. Support cross-layer monitoring—with defined metrics, thresholds, and protocols—that surfaces emerging misalignments across the Representation, Learning, and Calibration layers over time	Continuously aggregate cross-layer signals (drift, subgroup outcome disparities, instability); trigger alerts by severity/cadence; maintain cross-layer dashboards Illustrative methods: cross-layer drift detectors; alert rules; fairness dashboards; service-level monitors	Interpret escalations; set intervention scope/cadence, and stakeholder routing for high-stakes cases; define approval gates and documentation requirements	Execute fairness (re)calibration across affected layers; synchronize timing to avoid race conditions; stage rollouts/backouts; register post-change monitoring thresholds Illustrative methods: synchronized threshold/parameter updates; coordinated cadence protocols; staged deployment/backout plans	Approve cross-layer interventions; confirm (i) scope/cadence are proportionate, (ii) stakeholder routing and sign-offs are complete, and (iii) alignment with strategy, risk limits, and regulatory commitments
MR-CL2. Mitigate error propagation across layers by enabling cross-layer impact assessment (pre/post change) and interface–contract verification at layer handoffs.	Run pre-change simulations and post-change checks; verify schemas/encodings, units/scales and time windows, score type/calibration status, and dependency versions at handoffs Illustrative methods: schema diff and validator; unit/scale/time-window checks; calibration-status validator; dependency graph checks; acceptance test suites	Judge materiality and sequence of remediation; decide whether coordinated multilayer fixes are required and in what order; specify acceptance criteria for go/no-go	Generate remediation plan (e.g., staged data/model/policy updates); implement fixes; rerun impact checks and acceptance tests; prepare rollback manifests Illustrative methods: staged remediation; canary/baseline comparisons; rollback-readiness checks	Approve remediation and sequencing; verify (i) all post-change acceptance tests passed, (ii) no unresolved interface breaks remain, and (iii) rollback readiness is documented
MR-CL3. Maintain a unified, immutable provenance and approval-and-	Create signed, timestamped entries linking who/what/when/why to	Verify completeness/consistency of entries; reject records lacking rationale or evidence; enforce	Integrate logging instrumentation into continuous integration/deployment	Certify audit readiness and compliance; confirm (i) completeness/

authorization log for all fairness-relevant modifications to data (Representation), model (Learning), and decision policies (Calibration); support cross-layer traceability, versioning, and audit-ready reporting	evidence (simulations, impact assessments); ensure lineage continuity across layers Illustrative methods: automated logging instrumentation in pipelines; evidence attachment; lineage IDs; retention checks	retention and access policies; resolve conflicts in lineage	pipeline; enforce version linkage and rollback readiness; generate audit-ready reports Illustrative methods: Continuous integration/deployment gates; version linkage; rollback manifests; audit report generators	consistency of entries, (ii) evidence sufficiency for independent review, and (iii) retention/access controls per policy
Note: In routine settings, phases often follow a linear flow—AI agents (Monitor & Analyze) → Human agents (Contextualize & Validate) → AI agents (Generate & Refine) → Human agents (Adjudicate & Approve). In complex or high-stakes cases, phases iterate with deeper contextualization, rigorous ethical scrutiny, and repeated human–AI agent interactions. All approvals are recorded in the unified provenance and approval-and-authorization log.				

5.5. Governance of Human–AI Agent Collaboration Across the Decision-Task Portfolio

Building on the portfolio-level GRs (Table 3), we detail a unified risk-tiered federated governance mechanism for human–AI agent collaboration across the decision-task portfolio. It is enabled by two fairness complements: human skills (which enable the federated structure) and structured stakeholder feedback (which informs the risk-tiered AI agent autonomy policy). Together, this design enables fairness adaptation to scale across heterogeneous decision tasks, preserving local flexibility while maintaining global coherence.

5.5.1. Federated Governance

A federated structure (GR-1), anchored by a central fairness council and local decision-task councils, addresses the centralization-autonomy tension: centralized accountability and risk mitigation versus local flexibility. The central body establishes global minimum process standards (e.g., oversight protocols, evidence packs, escalation routes) for adaptive cycles, which are continuously informed by feedback from local councils. These standards define the baseline for human-AI agent collaboration, distinguishing non-negotiable portfolio requirements from configurable choices. Local councils implement these standards, negotiating those configurable choices to adapt protocols to their contexts, ensuring policy alignment while accommodating diverse task needs. This structure is enabled by a crucial fairness complement—the human skills to collaborate with AI agents, interpret AI outputs, and approve interventions.

5.5.2. Risk-Tiered AI Agent Autonomy

Within the federated structure, the risk-tiered AI agent autonomy policy (GR-2)—defining levels like automatic action or post hoc review—is determined by task risk and human skills, and updated based on reliability metrics. This policy governs the collaboration to temper AI agent autonomy with necessary human control, calibrating the balance across the portfolio by adjusting evidence criteria, approval thresholds, and decision rights. The policy is made responsive by its essential complement: structured stakeholder feedback (GR-3). This feedback—via inclusive forums like “fairness dialogues”—uncovers issues and calibrates AI agent autonomy bounds. Flowing upward to refine central protocols and downward to tailor local mechanisms, it maintains alignment between governance and frontline realities.

Counterfactually, without federated structure, centralized-only governance is rigid, whereas purely local governance fragments practices. Without risk-tiered autonomy, subunits risk either overly rigid or excessive autonomy; without stakeholder feedback, global protocols drift from local realities, missing context. Without human skills, AI diagnostics and interventions cannot be vetted or authorized, rendering collaboration ineffective. Accordingly, we advance:

Design Principle 5 (Governance of Human–AI Agents Across the Decision-Task Portfolio)

- Federated governance enabled by human skills.** Establish a federated structure where a central council defines global minimum process standards for adaptive cycles. Local councils negotiate configurable choices to tailor protocols to decision tasks—specifying evidence criteria, approval thresholds, and escalation. Human skills enable this structure by providing the capability to vet AI agent outputs and authorize fairness adaptations. Counterfactually, centralized governance is rigid; purely local governance fragments practices; and without skills, AI agent diagnostics and interventions cannot be vetted, rendering collaboration ineffective.
- Risk-tiered AI agent autonomy enabled by stakeholder feedback.** Define risk tiers that determine autonomy levels (e.g., automatic, preapproval, post hoc review) per task; update tiers based on reliability metrics and

structured stakeholder feedback on fairness to surface emergent risks and calibrate collaboration bounds. Counterfactually, static autonomy stalls responsiveness or inflates risk; absent feedback, autonomy drifts from ethical and operational realities.

Illustrative example—ED triage (Portfolio-level governance): In a large hospital system, departments balance human control and AI agent autonomy—some emphasize expert judgment augmented by AI insight; others, AI execution with oversight. Absent a federated structure that evolves global minimum standards, departments risk inconsistent practices (e.g., divergent definitions of “urgency”). Likewise, neglecting input from nurses or patient advocates exacerbates control–autonomy tensions, excluding frontline perspectives. With a federated structure enabled by human skills, and a risk-tiered policy informed by stakeholders, hospitals harmonize AI agent autonomy with human oversight and judgment in fairness adaptation.

6. Unifying Theoretical Proposition

FAIR advances a unifying theoretical proposition regarding organizational capability development dynamics: organizations can sustain fairness alignment and stakeholder trust through layer-specific and cross-layer adaptive cycles of AI artifacts, yet the magnitude and timing of these gains depend on governance structures, intangible fairness complements, and AI agent autonomy adjustments within workflows. Drawing on the managing-with-AI literature (Abbasi et al., 2024; Grote et al., 2024; Hillebrand et al., 2025; Rai et al., 2019), as well as insights on intangible investments (Brynjolfsson et al., 2021) and transformation of work (Brynjolfsson & Mitchell, 2017), we anticipate a short-run lag followed by acceleration in fairness outcomes.

This lag arises because adaptive cycles require recalibrating control–autonomy thresholds as contexts evolve. Excessive AI agent autonomy risks unaccountability; excessive human control limits prediction and learning. Whereas layer modularity underpins long-term gains, impact depends on how this autonomy is managed via risk-tiered federated governance (Asatiani et al., 2021; Grote et al., 2024). Such structures are crucial for clarifying accountability and bridging incentives between developers who seek speedy deployment and users who seek robust risk safeguards (Cobbe et al., 2023; Hillebrand et al., 2025).

Importantly, FAIR contends that this dynamic governance capability incurs learning costs before yielding benefits. Establishing federated bodies, conducting “fairness dialogues,” and building human skills to work with AI agents are deep, intangible investments (Brynjolfsson et al., 2021). They initially obscure near-term benefits but foster the necessary synergy for future gains. Absent these investments, local fixes may stall or backfire due to misalignment across layers or lack of stakeholder buy-in regarding AI agent autonomy. This leads to the following:

Proposition: *Organizations that adopt and enact FAIR’s sociotechnical design to develop reinforcing organizational capabilities—(i) artifact-level adaptation cycles executed by human–AI agent collaboration, (ii) federated governance of the decision-task portfolio enabled by human skills to work with AI agents, and (iii) risk-tiered AI agent autonomy made responsive by stakeholder feedback on fairness experiences—will initially yield limited or even negative net fairness improvements. As these sociotechnical mechanisms are learned, stabilized, and integrated, they ultimately yield a pronounced J-curve in fairness alignment and stakeholder trust.*

7. FAIR as a Design Theory

As detailed in Table 8, FAIR instantiates the components of a design theory (Gregor & Jones, 2007), integrating both explanatory and prescriptive knowledge for managing paradoxical fairness tensions in AI-automated decision-making. Its *purpose* is the ongoing, responsive adaptation of AI artifacts that automate decisions; its *scope* is AI-automated decision-making in organizations (not purely individual use or settings lacking organizational governance).

FAIR’s *constructs* include: (i) metatheoretical fairness dimensions—principles (distributive, procedural, interactional), goals (fairness, predictive accuracy, operational efficiency, economic), and foci (individual, group, societal); (ii) system requirements covering both artifact-level MRs within and across layers—Representation, Learning, Calibration—and portfolio-level GRs; and (iii) sociotechnical design mechanisms, encompassing artifact-level adaptive cycles and portfolio-level risk-tiered federated governance of decision tasks.

FAIR addresses *principles of form and function* by distinguishing AI artifacts (data, models, and decision policies) from the agents who adapt them: AI agents (monitor/analyze, generate/refine) and human agents (contextualize/validate, adjudicate/approve). This distinction aligns with responsible-AI design emphasizing transparency, interpretability, and adaptive governance (Abbasi et al., 2024). The artifact is organized as a modular pipeline—Representation (data), Learning (model), and Calibration (decision)—with fairness refinements implemented through Surfacing and Resolving cycles.

Artifact mutability follows from this modular architecture, which enables incremental, layer-specific adaptations of the AI artifact without destabilizing the entire system and maintains responsiveness without large-scale disruptive changes.

FAIR's *justificatory knowledge* for the theory of the problem integrates the paradox lens (Smith & Lewis, 2011) with multitheoretical fairness insights (Ethics, Organizational Justice, Economic Fairness, Rawlsian Justice). The *justificatory knowledge* for the theory of the solution draws on the paradox lens (Raisch et al., 2018), AI/ML advances (layered modularity, computational fairness techniques), and the emergent literature on managing-with-AI (Abbasi et al., 2024; Grote et al., 2024; Hillebrand et al., 2025; Murray et al., 2021; Rai et al., 2019).

FAIR's *implementation principles* clarify how organizations differentiate between sequential and iterative Surfacing and Resolving phases in fairness adaptation, varying human and AI agent roles by task stakes. This is operationalized through risk-tiered federated governance. The federated structure is enabled by human skills that **support** local flexibility and global coherence, while structured stakeholder feedback makes AI autonomy responsive to context, calibrating the balance between human control and AI autonomy to situational demands.

FAIR's design culminates in a *testable proposition*. The essence of this proposition is that its two reinforcing mechanisms—artifact-level adaptation and portfolio-level risk-adjusted federated governance—are deep, intangible investments. Positing that organizations enact these mechanisms, enabled by human skills and stakeholder feedback on their fairness experiences, will initially cause a lag in fairness gains. FAIR predicts that as they are learned, stabilized and integrated, they yield J-curve returns in fairness alignment and stakeholder trust.

Finally, an *expository instantiation* in an ED triage setting shows how fairness tensions across AI artifact layers are surfaced and resolved through human-AI agent collaboration, with human control and AI agent autonomy calibrated to risk tiers of decision tasks.

Table 8. FAIR as a Design Theory	
Component	FAIR Operationalization
Purpose	Manage paradoxical fairness tensions in AI-automated decision-making
Scope	AI-automated decision-making in organizations
Constructs	<i>Metatheoretical fairness dimensions</i>: principles, goals, and foci <i>System requirements</i>: (i) artifact-level MRs: layer-specific (Representation, Learning, Calibration), plus cross-layer; and (ii) portfolio-level requirements for governance of AI-automated tasks <i>Sociotechnical design mechanisms</i>: (i) Artifact-level: Adaptive cycles executed collaboratively by human and AI agents; (ii) Portfolio-level: risk-tiered federated governance
Principles of Form & Function	- Modular, layered AI artifact design enables targeted monitoring, diagnosing, and intervention - Distinguishes AI artifacts that automate decisions from the agents who adapt them: AI agents (monitor/analyze, generate/refine) and human agents (contextualize/validate, adjudicate/approve)
Artifact Mutability	Enables incremental, modular fairness adaptations at each layer with cross-layer coordination; maintains responsiveness without large-scale disruptive changes
Justificatory Knowledge	<i>Theory of the Problem</i>: Paradox lens; fairness theories (Ethics, Organizational Justice, Economic Fairness, Rawlsian Justice) <i>Theory of the Solution</i>: Paradox lens; AI/ML advances (layered, modular AI artifact pipeline; computational fairness); human-AI agent augmentation frameworks; managing-with-AI literature
Principles of Implementation	- AI agent autonomy and human control vary by task risk - Federated governance enabled by human skills supports global coherence and local flexibility - Risk-tiered AI agent autonomy made responsive by stakeholder feedback calibrates control to task risk
Testable Propositions	Short- and long-term impacts (J-curve) on fairness alignment and stakeholder trust from (i) artifact-level adaptation cycles executed by human-AI agent collaboration, (ii) portfolio-level federated governance of decision tasks enabled by human skills to work with AI agents, and (iii) risk-tiered AI agent autonomy made responsive by stakeholder feedback on fairness experiences
Expository Instantiation	ED triage: Surfacing and Resolving cycles enacted at each artifact layer, plus cross-layer, collaboratively by human and AI agents; human control-AI agent autonomy calibrated to decision-task risk

8. Implications

8.1. Theoretical Implications

FAIR's primary theoretical implication is its reframing of AI fairness as both a problem and solution. First, it provides a theory of the problem that moves beyond static critiques (Dolata et al., 2022; Rana et al., 2022) to explain why fairness tensions persist in AI-automated decision-making. It does so by recasting AI fairness as an in-

operation design problem (Cowgill & Tucker, 2020) and conceptualizing its tensions as paradoxical—persistent, interdependent, and dynamically evolving (Smith & Lewis, 2011). Second, this diagnosis necessitates FAIR’s theory of the solution: a sociotechnical design, operationalized by two reinforcing mechanisms.

This theory of the problem contributes to the literature in three ways. First, by conceptualizing AI fairness as a paradox, we challenge static, reactive approaches and provide a unifying typology (constraints, incompatibilities, trade-offs, dilemmas) to organize disparate research on fairness tensions. Second, by synthesizing four theoretical perspectives (Ethics, Organizational Justice, Economic Fairness, Rawlsian Justice), we move beyond a single-theory lens to identify three underlying metatheoretical dimensions (principles, goals, foci). Third, we posit that the interdependencies within and among these dimensions are the root, endogenous source that constitutes the AI fairness paradox, explaining its persistence.

FAIR’s theory of the solution proposes a design theory operating at two linked levels: (i) artifact-level adaptive cycles and (ii) portfolio-level risk-tiered federated governance.

At the artifact level, FAIR’s first contribution is its modular, layered design (Representation, Learning, Calibration)—the “what” of the system. This conceptualization introduces a separation of concerns to AI fairness, a classic design principle. It provides a novel prescription for how specific paradoxes (e.g., trade-offs, incompatibilities) can be surfaced and resolved at distinct artifact layers. Cross-layer coordination manages the interdependencies that modularity alone cannot eliminate, preventing myopic, single-layer fixes.

This layered artifact is then acted upon by FAIR’s second artifact-level contribution: the operationalized sociotechnical mechanism that defines the “who” (the agents) and “how” (the adaptive cycles) for managing the paradox. This mechanism extends the managing-with-AI literature (Abbasi et al., 2024; Grote et al., 2024; Hillebrand et al., 2025; Karunakaran et al., 2025; Rai et al., 2019) by explicitly designing human–AI agent collaboration to responsibly derive complementarities. It moves beyond high-level principles to specify two reinforcing cycles for surfacing and resolving fairness tensions: Surfacing (where AI agents monitor/analyze and human agents contextualize/validate) and Resolving (where AI agents generate/refine and human agents adjudicate/approve). This interaction, ranging from sequential to iterative based on task risk, provides a concrete design for fairness adaptation of AI artifacts through human–AI agent collaboration.

At the portfolio level, the risk-tiered federated governance mechanism challenges static, one-size-fits-all governance models. By extending work on federated AI governance (Asatiani et al., 2021; Grote et al., 2024), FAIR provides a design for the dynamic calibration of human control and AI agent autonomy, aligning it with task risk—from high supervision in health-care triage to AI autonomy in low-stakes tasks (Raisch & Krakowski, 2021).

A key implication of FAIR is bridging the AI fairness literature with the economic theory of intangible investments (Brynjolfsson & Mitchell, 2017; Brynjolfsson et al., 2021). FAIR conceptualizes the complements of human skills and stakeholder feedback as deep, intangible investments. Necessary human skills enable the federated structure to function, while structured stakeholder feedback makes the risk-tiered policy responsive. By reframing these complements as investments, FAIR provides a novel lens to explain why organizations will likely incur near-term learning costs, creating a J-curve effect in fairness alignment and stakeholder trust.

Ultimately, FAIR lands on a core premise. Its theory of the problem provides an integrated diagnosis of AI fairness as a paradox, while its theory of the solution specifies a novel design for **F**airness **A**daptation (the *what*—the continuous, layered adaptation of AI artifacts) through **A**I-augmented **R**esponsiveness (the *how*—the reinforcing sociotechnical mechanisms of human–AI agent collaboration at the artifact level and federated governance at the decision-task portfolio level).

8.2. Practical Implications

FAIR’s core implication is a fundamental shift in managerial mindset: fairness is a continuous adaptation of AI artifacts that automate decisions, not a one-off fix. As automation scales, managers also need to shift the frame from isolated artifacts to a portfolio-wide approach.

In diagnosing fairness issues, FAIR requires managers to apply a stakeholder lens to fairness experiences, surfacing the paradoxical tensions within and among principles, goals, and foci. This diagnostic clarity allows managers to move beyond static and siloed metrics, continually adapting AI decision protocols to resonate with stakeholder values and sustain trust.

Because AI systems operate at speed and scale, fairness adaptation must be proactive. FAIR provides this via human–AI agent collaboration to manage paradoxical tensions within and across layers of the AI artifact pipeline. When applied, this design delineates complementary roles in Surfacing (where AI agents monitor and analyze, while humans contextualize and validate) and Resolving (where AI agents generate and refine, while humans adjudicate and approve)—to ensure adaptation is both scalable and socially accountable.

Organizations need to proactively harmonize global coherence and local flexibility of fairness adaptation across their AI-automated decision-task portfolio. By using risk-tiered federated governance, they can establish the structures and processes to define global minimum process standards and negotiate decision-task-specific configurable choices for human–AI agent collaboration, as well as calibrate the balance of human control and AI agent autonomy to the task risk.

Finally, the effectiveness of FAIR’s design—both the human–AI agent collaboration and risk-tiered federated governance—hinges on investing in two fairness complements. The first, and most critical, is human skills to work with AI agents. This capability must be developed throughout the organization, enabling frontline agents to adapt artifacts and governance councils to harmonize global coherence and local flexibility in fairness adaptation. The second is structured stakeholder feedback, which guides the risk-tiering that calibrates AI agent autonomy versus human control for decision tasks.

Collectively, this sociotechnical design—integrating continuous fairness adaptation at the artifact level with a risk-tiered governance for the decision-task portfolio—builds the organizational capability for responsive, coherent, and trustworthy AI fairness.

8.3. Future Research Implications

By conceptualizing AI fairness as a sociotechnical paradox, FAIR shifts attention to persistent, interdependent tensions that evolve with AI capabilities, organizational priorities, stakeholder expectations, and regulation. Researchers can investigate how organizations frame and manage these tensions and how triggers, such as new regulation or stakeholder demands, reshape the interdependencies within and among principles, goals, and foci.

FAIR’s interventions in the AI pipeline—Representation (data), Learning (model), and Calibration (decision)—raise questions about how AI artifacts become “theory-laden.” Researchers can examine how fairness lenses (Ethics, Organizational Justice, Economic Fairness, Rawlsian Justice) (i) translate into metrics at the Representation layer, (ii) guide trade-off or joint optimization at the Learning layer, and (iii) shape policy thresholds at the Calibration layer. This will help uncover how assumptions are encoded in artifact design and reconfigure trade-offs across decision risk (e.g., high-stakes ED triage vs. low-stakes tasks).

A promising avenue concerns transitions in levels of AI agent autonomy. Low-stakes tasks may rely on AI actions with post hoc human review, whereas high-stakes decisions demand greater human oversight. Underexplored questions include which capabilities enable shifts from human-led to AI-led processes (and back), how AI agent autonomy is scaled back when fairness issues emerge, and how human and AI agents (e.g., bias-detection vs. interpretability roles) collaborate or conflict during fairness adaptations. Finally, exploring iterative refinement loops—feed-forward simulations and feedback-based recalibrations across layers—would show how inputs evolve to produce scalable, context-sensitive adaptations.

FAIR proposes federated governance as a middle ground between centralization and decentralization, underpinned by fairness complements—human skills to collaborate with AI agents and structured stakeholder feedback on fairness in AI-automated decision tasks. Researchers can examine how introducing interpretability toolkits for human agents working with AI agents may spark demand for specialized training of human agents and prompt revisions to human–AI agent collaboration protocols. Longitudinal research can examine how localized fairness adaptations accumulate over time, reshaping frontline workflows and feeding back into portfolio-level governance through revised standards, risk tiers, and escalation protocols.

Scholars can also investigate how adversarial and deliberative viewpoints are integrated in fairness adaptation—for example, through stakeholder panels that stress-test fairness claims and adversarial AI agents that generate counter-evidence and counter-arguments—to strengthen the robustness and validity of fairness adaptation. In addition, researchers can examine how the design of stakeholder feedback architectures allocates voice, mitigates power asymmetries, and shapes which fairness signals are surfaced, amplified, or ignored—dynamics that are crucial to sustain organizational capabilities for fairness adaptation in AI artifacts that automate decision tasks.

Finally, as generative AI advances, organizations face new fairness opportunities and vulnerabilities, including emergent biases in large pretrained models. Moreover, as AI is increasingly componentized, FAIR’s sociotechnical design provides a scaffold to monitor, calibrate, and adapt decision-making components embedded within assemblages to maintain fairness. Analyzing feedback loops across operational and policy layers can show how risk-tiered federated governance and structured stakeholder feedback shape legitimacy, stakeholder trust, and resilience. FAIR’s core principles—treating fairness as a sociotechnical paradox, using modular interventions, and balancing AI agent autonomy with human control—provide a foundation. This allows IS scholars to develop insights that keep AI systems ethically grounded and socially accountable amid rapid technological change.

9. Conclusion

FAIR reframes AI fairness from a static, technical problem to a sociotechnical paradox, requiring proactive, adaptive management rather than a one-off solution. It identifies the root source of this paradox as the interdependencies within and among metatheoretical dimensions—principles, goals, and foci—that cut across individual fairness lenses.

To manage this paradox, FAIR operationalizes two reinforcing sociotechnical mechanisms: (i) at the artifact level, adaptive cycles (Surfacing/Resolving) executed by human–AI agent collaboration, and (ii) at the portfolio level, a unified risk-tiered federated governance that supports coherence across decision tasks while providing flexibility to individual tasks. The federated structure is enabled by the complement of human skills to work with AI agents, whereas the risk-tiered AI agent autonomy policy is made responsive by the complement of structured stakeholder feedback on fairness experiences. Together, these mechanisms build an organizational capability for AI-automated decision-making to remain fair, responsive, ethically grounded, and accountable—providing a foundation for sustained fairness alignment and stakeholder trust.

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Appendix

Types of AI Fairness Tensions: A Synthesis of IS Research

By synthesizing insights from 16 AI fairness studies published in leading IS outlets (*Information Systems Research*, *MIS Quarterly*, *Journal of Management Information Systems*, *Journal of the Association for Information Systems*, *Decision Support Systems*, *European Journal of Information Systems*, *Information Systems Journal*, and *Management Science*) between 2010-2025, we identify four distinct yet interconnected types of fairness tensions: *constraints*, *trade-offs*, *incompatibilities*, and *dilemmas*.

Fairness tension as constraint: Fairness often takes the form of nonnegotiable boundaries in regulation, societal expectations, and organizational policy, which impose conditions on AI systems. Frameworks such as GDPR and the EU AI Act restrict algorithmic practices—promoting fairness and accountability yet complicating bias detection (Kordzadeh & Ghasemaghahi, 2022; Xu & Zhang, 2021). Legal protections (e.g., the Fair Housing Act) require firms like Facebook to enforce fairness criteria in ad targeting (Teodorescu et al., 2021). Constraints can degrade accuracy—such as removal of gender from hiring models (Dargnies et al., 2026)—and equal-treatment compliance can worsen disparate impacts (Fu et al., 2021). Beyond regulation, societal expectations impose implicit constraints (Dolata et al., 2022; Kallus et al., 2021). Organizational policies do likewise, such as a two-stage target-debt framework for online ad allocation with service-level fairness requirements (Li et al., 2025).

Fairness tension as trade-off (zero-sum): Trade-offs arise among fairness goals, operational objectives, and strategy. Within the organization, fairness definitions can conflict: optimizing equal opportunity may discourage effort, disadvantaging all groups relative to equal treatment (Fu et al., 2022); tensions among fairness, accountability, and transparency mean fairness can reduce transparency or accountability (Zhdanov et al., 2022). Operationally, fairness often trades off with model performance: fairness gains in credit scoring can lower predictive accuracy, requiring balancing (Hurlin et al., 2026). Strategically, fairness may misalign with efficiency, profitability, or welfare: efficiency-optimized mechanisms can be unfair (Bichler et al., 2021); fairness constraints can raise or reduce welfare depending on stringency (Cohen et al., 2022); and rigorous validation for ethical alignment can slow artifact iteration (Abbasi et al., 2024).

Fairness tension as incompatible assumptions (winner-takes-all): Irreconcilable ethical assumptions embedded in fairness definitions create *absolute* incompatibilities between consequentialism (outcomes) and deontology (rules). Algorithmic mechanisms cannot simultaneously achieve efficiency (consequentialist) and envy-free outcomes (deontological), forcing ethical prioritization (Bichler et al., 2021). Equal treatment (deontological) and equal opportunity (consequentialist) also conflict—uniform standards vs. differentiated treatment to reach equitable outcomes (Fu et al., 2022). Computational work shows that multiple fairness goals often cannot be jointly achieved when groups differ systematically (Bichler et al., 2021; Cohen et al., 2022), and demographic parity can conflict with equal opportunity (Fu et al., 2022; Hurlin et al., 2026). Amazon’s hiring algorithm exemplifies such clashes: efficiency-oriented learning disadvantaged women, violating nondiscrimination norms (Dastin, 2022).

Fairness tension as dilemma (no-win): Dilemmas arise when any decision causes harm, forcing a choice between negative outcomes. Although disclosing racial disparities of an AI system raises human awareness, it can complicate labelers’ assessments of the system’s trustworthiness and increase inconsistency in their judgments (Rhue, 2023). Similarly, using impact parity to enforce fairness may benefit a protected group with higher acceptance rates, but it risks harming them in the long term by reducing overall algorithmic accuracy (Fu et al., 2022). These no-win situations require nuanced ethical reasoning and human adjudication.

Fairness tension as paradox: Although most researchers do not adopt a paradox lens, many imply the persistence, interdependence, and dynamism of tensions. Persistence appears in enduring debates and unresolved definitions (Hurlin et al., 2026; Teodorescu et al., 2021). Interdependence arises from incompatibilities—improving one criterion worsens another (Bichler et al., 2021). Dynamism follows from shifting societal norms, organizational priorities, and regulations, calling for adaptive strategies (Cohen et al., 2022; Teodorescu et al., 2021).

Summary: Our synthesis yields three insights. First, AI fairness tensions arise not only from exogenous sources (e.g., regulatory mandates) but also from endogenous interdependencies within and among principles, goals, and foci. Second, interdependencies complicate management: tensions trigger or morph into one another (e.g., removing

protected attributes can create accuracy trade-offs, amplifying incompatibilities such as individual vs. demographic fairness), and attempts to resolve incompatibilities can escalate into dilemmas where every option entails harm. Third, reactive and isolated “one-and-done” fixes are insufficient; proactive, adaptive strategies are needed to manage persistent, interdependent, dynamic tensions.